A Hamiltonian–Geometric Optimization Framework for Machine Learning

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Abstract

This manuscript develops and evaluates a Hamiltonian–geometric optimizer for machine learning. It presents (1) the mathematical background of the framework; (2) an analytical validation of the update equations against the underlying Hamiltonian mechanics; (3) empirical evidence from classical, geometric, and variational quantum benchmarks, together with visual diagnostics of the metric structure; and (4) an independently coded, paired-seed, bootstrap-and-Holm-corrected extended ablation study that replicates the central curvature-dependent finding under a second implementation while also reporting where a fixed, non-task-tuned setting of the full architecture underperforms simple baselines, on a chaotic quantum-control task and a GPU-scale classification task. A dedicated derivation shows how gradient descent, momentum, Adam/RMSProp-style diagonal preconditioning, natural-gradient-style methods, entropy descent, and the full Hamiltonian-geometric optimizer arise within a single update family; it also gives Adam's diagonal-metric update from a local variational step principle, and a further derivation extends the same update family to modern optimizers (AdamW, Lion, Shampoo/K-FAC, and Muon), each obtained by a specific choice of metric, kinetic term, or potential splitting rather than by analogy. The analysis also establishes the sign convention for the spectral Hamiltonian and supplies an explicit vector-to-matrix construction for the memory metric M_{ij}, yielding a self-contained formulation suitable for empirical comparison.

Keywords: Hamiltonian optimization; Riemannian metrics; adaptive optimizers; geometric machine learning; variational quantum algorithms.

1 Introduction

Optimization algorithms in machine learning can often be interpreted as discretized dynamical systems on parameter space. Gradient descent is the overdamped limit of a flat dissipative flow; momentum methods retain an inertial state; adaptive methods introduce coordinate-dependent scaling; and natural-gradient methods replace Euclidean geometry by a task-dependent metric. These viewpoints are usually presented separately, which obscures the common geometric structure shared by many optimizer families.

The Hamiltonian–geometric framework studied here treats parameters as generalized coordinates, the loss as potential energy, and the optimizer state as momentum evolving on a metric parameter manifold. This formulation gives a single update law containing flat-gradient methods, momentum, diagonal adaptive preconditioning, natural-gradient-style updates, entropy descent, and the full Hamiltonian-geometric optimizer as limiting or structured cases. The central question is therefore twofold: whether the formulation is internally consistent as a mathematical dynamical system, and whether the additional geometric structure yields measurable advantages on workloads where curvature, conditioning, or phase-space behavior are relevant.

The manuscript makes four contributions. First, it derives the unified update family and gives a variational derivation of Adam as a diagonal-metric local step. Second, it extends the same view to modern optimizers, including AdamW, Lion, Shampoo/K-FAC, AdaFactor, LARS/LAMB, and Muon. Third, it evaluates the method on classical learning tasks, physics-informed learning, geometric diagnostics, a chaotic kicked-top control problem, and QAOA MaxCut. Fourth, it reports an independently coded extended ablation study (§13) built specifically to stress-test the third contribution's findings under a second implementation, higher seed counts, and corrected multiple comparisons. The resulting evidence supports a bounded claim: Hamiltonian-geometric optimization is especially effective when full-metric structure is available, while remaining competitive rather than uniformly dominant against the strongest modern baselines.

2 Related Work

The dynamical-systems view of optimization is not new, and the contribution here must be positioned against it precisely rather than implied to be novel by omission. Su, Boyd & Candès [1] first modeled Nesterov's accelerated gradient method as the small-step-size limit of a second-order ODE, opening a line of work that reads discrete optimizers as discretizations of continuous-time dynamics. Wibisono, Wilson & Jordan [2] generalized this to a family of Bregman-Lagrangian dynamics whose Euler–Lagrange equations recover essentially all known accelerated first-order methods, already placing momentum and mirror-descent-style metrics in one variational framework. Betancourt, Jordan & Wilson [3] and Maddison et al. [4] – whose title gives "Hamiltonian descent" its name – go further and cast accelerated methods explicitly as (dissipative, symplectic, or conformal) Hamiltonian systems on parameter-momentum phase space, with França et al. [5] analyzing the conformal-symplectic integrators that make such dynamics stable under discretization – the same concern this report's own energy-backtracking safeguard (§10) addresses empirically rather than through a structured integrator. Natural gradient [11] and mirror descent [6] are older, narrower instances of the same idea: replace the Euclidean metric implicit in gradient descent with a task-dependent Riemannian or Bregman geometry. Riemannian and manifold optimization [7] develops optimization directly on curved manifolds using retractions and parallel transport; this report's $g_{ij}(\theta)$ and its curvature correction F_{geo} (Eq. (5)) are closely related to that literature's Christoffel-symbol corrections for a moving metric, a connection worth making precise in future work rather than only qualitatively here. A distinction that must also be drawn explicitly: Hamiltonian Monte Carlo [8] uses the identical coordinate/momentum/Hamiltonian machinery, but for *sampling* from a target distribution via conservative (energy-preserving) trajectories: what distinguishes the present, optimization-oriented use is precisely the dissipative modification (Rayleigh damping, Eq. (6)) that HMC deliberately avoids in order to preserve the target measure.

What is not already established in this literature, and is the actual contribution tested here, is threefold. First, the prior work above is almost entirely a *descriptive* theory of existing accelerated methods (showing that Nesterov, heavy-ball, and mirror descent already are instances of a continuous-time dynamical system); this report instead treats the dynamical system as a *prescriptive* template for a new, directly trainable optimizer with a metric built from the loss's own regularized Hessian, an explicit non-Markovian memory force, and a spectral-entropy regularizer on the metric's own eigenvalue spread – none of which appear as a package in the works above. The same template also recovers Adam and RMSProp themselves as a diagonal-metric special case (§4), rather than only the momentum-based accelerated methods the ODE-limit literature above already covers. Second, §5 extends the same correspondence to optimizers published independently and after this line of theoretical work (AdamW, Lion, Shampoo, Muon), which to our knowledge have not been placed on a single metric/kinetic-term/potential-splitting axis elsewhere. Third, §9–§9.4 subject the resulting optimizer to the same empirical stress-testing (multi-seed variance, width scaling, ablation of each additive term) that a purely theoretical ODE-limit analysis does not by itself provide, and report the negative and mixed results (§9.4, Table 14) alongside the favorable ones.

3 Mathematical Background

3.1 Coordinates, potential, and the Hamiltonian

The framework treats model parameters $\theta = (\theta^1, \ldots, \theta^n)$ as generalized coordinates $q^i \equiv \theta^i$, with the loss function playing the role of potential energy, $V(q) \equiv L(\theta)$. Given a metric $g_{ij}(\theta)$ on parameter space defining a kinetic term

$$T = \tfrac{1}{2}\,\dot{\theta}^i\, g_{ij}(\theta)\,\dot{\theta}^j, \tag{1}$$

the Legendre transform of the Lagrangian $\mathcal{L} = T - V$ gives canonical momentum $p_i = \partial\mathcal{L}/\partial\dot{\theta}^i = g_{ij}\dot{\theta}^j$ and Hamiltonian

$$H(\theta, p) = \tfrac{1}{2}\,p_i\, g^{ij}(\theta)\, p_j + L(\theta), \tag{2}$$

where g^{ij} is the matrix inverse of g_{ij}, $g^{ij}g_{jk} = \delta^i_k$.

3.2 Hamilton's equations and the geometric force

Hamilton's equations $\dot{\theta}^i = \partial H/\partial p_i$, $\dot{p}_i = -\partial H/\partial\theta^i$ applied to Eq. (2) give

$$\dot{\theta}^i = g^{ij}(\theta)\, p_j, \tag{3}$$

$$\dot{p}_i = -\partial_i L - \tfrac{1}{2}\, p_j p_k\, \partial_i g^{jk}. \tag{4}$$

The second term in Eq. (4) is named the *geometric force*,

$$F_{\text{geo}} \;=\; \tfrac{1}{2}\, p^{\mathsf{T}} \big(\nabla_\theta g^{-1}\big)\, p, \tag{5}$$

so that Eq. (4) reads $\dot{p}_i = -\partial_i L - F_{\text{geo}}$: motion is driven by the loss gradient *and* by how the metric itself curves across parameter space.

3.3 Dissipation

Machine-learning optimization is not conservative – the loss must decrease – so a Rayleigh dissipation function $\mathcal{R}(p) = \frac{1}{2}\gamma\, p^i p_i$ is added, giving $\dot{p}_i = -\partial H/\partial\theta^i - \partial\mathcal{R}/\partial p_i$, i.e.

$$\dot{\theta}^i = g^{ij}(\theta)\, p_j, \qquad \dot{p}_i = -\partial_i L - F_{\text{geo}} - \gamma p_i. \tag{6}$$

In the flat Euclidean case ($g_{ij} = \delta_{ij}$, so $\partial_i g^{jk} = 0$ identically and $F_{\text{geo}} = 0$), Eq. (6) reduces to $\dot{\theta} = p$, $\dot{p} = -\nabla_\theta L - \gamma p$, and in the overdamped limit ($p \approx -\gamma^{-1}\nabla_\theta L$) this Euler-discretizes to plain gradient descent. Classical momentum and Adam are presented as further special cases: momentum as damped Hamiltonian flow with $p_t \sim v_t$; Adam as an approximate *diagonal*-metric Hamiltonian optimizer, $g_t^{ij} \approx \delta^{ij}/(\sqrt{v_t^i} + \epsilon)$, with the explicit caveat that Adam has no analogue of F_{geo} since v_t is treated as locally constant.

3.4 The generalized update rule

With a memory (non-Markovian) correction built from an exponentially weighted kernel over past gradients,

$$F_{\text{mem}}(\theta_t) = \mu \sum_{k=1}^{t} \kappa^{\,t-k}\, \nabla_\theta L(\theta_k), \qquad 0 < \kappa < 1, \tag{7}$$

the proposed discretized optimizer is

$$p_{t+1} = \beta p_t - \eta\Big[\nabla_\theta L + F_{\text{geo}} + F_{\text{mem}} + \alpha\nabla_\theta S(g)\Big], \tag{8}$$

$$\theta_{t+1} = \theta_t + \eta\, g^{-1}\, p_{t+1}. \tag{9}$$

The metric itself may be built from the (regularized) Hessian of the loss, $g_{ij} = H_{ij} + \lambda\delta_{ij}$, $H_{ij} = \partial^2 L/\partial\theta^i\partial\theta^j$, $\lambda > 0$.

Table 1: Optimizer families as limits or approximations of the Hamiltonian–geometric update.

Optimizer family	Hamiltonian specialization	What is removed or approximated
Gradient descent	Flat metric, overdamped limit	Momentum relaxes instantly; $F_{\text{geo}} = 0$
Heavy-ball momentum	Flat metric with finite momentum	No curvature, memory, or spectral force
Nesterov-style momentum	Flat damped flow with lookahead gradient	Lookahead discretization of momentum flow
RMSProp / Adam	Diagonal adaptive metric approximation	Ignores F_{geo} from metric variation
Natural gradient	Full information metric	Usually no Hamiltonian momentum or memory force
Entropy descent	Full/preconditioned metric step	Keeps g^{-1} but drops momentum and memory
Hamiltonian-geometric	Full metric with optional forces	Retains momentum, metric, memory, entropy, curvature

Memory-metric term M_{ij}

The metric extension $g_{ij} = H_{ij} + \mu M_{ij}$ requires M_{ij} to be a rank-2 symmetric object, while Eq. (7) defines F_{mem} as a vector force with one free index. A direct vector-to-matrix lift is therefore needed before the memory contribution can enter the metric. Use the same exponentially weighted construction Eq. (7) already uses, applied one power higher, to the *outer product* of the gradient with itself rather than to the gradient alone:

$$M_{ij}(\theta_t) = \kappa\, M_{ij}(\theta_{t-1}) + \big(\partial_i L(\theta_t)\big)\big(\partial_j L(\theta_t)\big), \tag{10}$$

with the same decay κ as Eq. (7) (so no new hyperparameter is introduced) and $M_{ij}(\theta_0) = 0$. This is exactly the uncentered empirical-Fisher construction used to motivate Adam's diagonal metric below (§4, "RMSProp and Adam") ($v_t \approx$ diag of an EMA of squared gradients) – Eq. (10) is the same recursion kept as a full matrix instead of truncated to its diagonal. Three properties make it a legitimate metric contribution rather than an arbitrary patch: (1) it is manifestly symmetric, $M_{ij} = M_{ji}$, since $(\partial_i L)(\partial_j L) = (\partial_j L)(\partial_i L)$; (2) it is positive semidefinite for any gradient sequence, since it is a nonnegative sum of rank-1 PSD terms vv^{T}, so $g_{ij} = H_{ij} + \lambda\delta_{ij} + \mu M_{ij}$ can only add curvature, never break the positive-definiteness already established for H; and (3) it reduces to exactly F_{mem}'s own recursion in the one-dimensional case, $M(\theta_t) = \kappa M(\theta_{t-1}) + (\partial L)^2$, matching Eq. (7) squared term-by-term. Thus the memory metric is not an additional free structure: it is the natural rank-2 lift of the same exponential-memory law already present in the vector force.

4 Standard Optimizers as Special Cases

The unifying claim is structural: standard optimizer families appear by choosing particular metrics, damping limits, and force terms in the Hamiltonian–geometric update. The master form is

$$p_{t+1} = \beta p_t - \eta \left[\nabla_\theta L + F_{\text{geo}} + F_{\text{mem}} + \alpha \nabla_\theta S(g)\right], \tag{11}$$
$$\theta_{t+1} = \theta_t + \eta g^{-1}(\theta_t) p_{t+1}. \tag{12}$$

Different optimizers are obtained by deleting or restricting pieces of this system:

Gradient descent

Set the metric to the Euclidean metric, $g = I$, remove memory and spectral forces, and note that $\nabla_\theta g^{-1} = 0$, so $F_{\text{geo}} = 0$. The dissipative continuous-time Hamiltonian system reduces to

$$\dot\theta = p, \qquad \dot p = -\nabla_\theta L - \gamma p. \tag{13}$$

In the overdamped limit, momentum equilibrates much faster than position: $\dot p \approx 0$, hence

$$p \approx -\gamma^{-1}\nabla_\theta L. \tag{14}$$

Substitution into $\dot\theta = p$ gives

$$\dot\theta \approx -\gamma^{-1}\nabla_\theta L. \tag{15}$$

Forward Euler with step size h therefore gives

$$\theta_{t+1} = \theta_t - \frac{h}{\gamma}\nabla_\theta L(\theta_t) = \theta_t - \eta\nabla_\theta L(\theta_t), \tag{16}$$

which is ordinary gradient descent.

Heavy-ball momentum

Keep the same flat metric $g = I$ and $F_{\text{geo}} = F_{\text{mem}} = \alpha\nabla S(g) = 0$, but do not take the overdamped limit. The master update becomes

$$p_{t+1} = \beta p_t - \eta\nabla_\theta L(\theta_t), \tag{17}$$
$$\theta_{t+1} = \theta_t + \eta p_{t+1}. \tag{18}$$

Define the usual descent velocity $v_t = -p_t$. Then

$$v_{t+1} = \beta v_t + \eta\nabla_\theta L(\theta_t), \tag{19}$$
$$\theta_{t+1} = \theta_t - \eta v_{t+1}. \tag{20}$$

Up to the conventional placement of the learning rate inside or outside the velocity variable, this is the heavy-ball momentum update.

Nesterov-style momentum

Nesterov momentum uses the same flat damped Hamiltonian skeleton but evaluates the gradient at a lookahead point. Starting from the heavy-ball reduction, replace the force evaluation point by

$$\tilde\theta_t = \theta_t - \beta v_t. \tag{21}$$

The update becomes

$$v_{t+1} = \beta v_t + \eta \nabla_\theta L(\tilde{\theta}_t), \tag{22}$$
$$\theta_{t+1} = \theta_t - \eta v_{t+1}. \tag{23}$$

Thus Nesterov is not a new metric; it is a lookahead discretization of the same flat dissipative Hamiltonian momentum flow.

RMSProp and Adam

RMSProp [10] and Adam [9]: let the metric be diagonal and time-adaptive:

$$g_t = \mathrm{diag}\left(\sqrt{\hat{v}_t} + \epsilon\right), \qquad g_t^{-1} = \mathrm{diag}\left((\sqrt{\hat{v}_t} + \epsilon)^{-1}\right), \tag{24}$$

where $\hat{v}_t$ is an exponential moving average of squared gradients. With momentum m_t identified with the descent momentum, the metric-preconditioned step is

$$\theta_{t+1} = \theta_t - \eta\, g_t^{-1} m_t = \theta_t - \eta\, \frac{m_t}{\sqrt{\hat{v}_t} + \epsilon}. \tag{25}$$

This is the Adam/RMSProp form. The correspondence is deliberately qualified: Adam treats g_t as locally fixed during the step, so it drops

$$F_{\text{geo}} = \tfrac{1}{2} p^\mathsf{T} (\nabla_\theta g^{-1}) p. \tag{26}$$

Therefore Adam is a diagonal-metric approximation to the Hamiltonian optimizer, not the full Hamiltonian-geometric optimizer.

The same Adam form can also be derived variationally from a local metric-weighted step principle. At iteration t, approximate the loss by its first-order expansion around θ_t and penalize motion using a positive diagonal metric G_t:

$$\Delta\theta_t = \arg\min_{\Delta} \left[m_t^\mathsf{T} \Delta + \frac{1}{2\eta} \Delta^\mathsf{T} G_t \Delta \right]. \tag{27}$$

Here m_t is the exponentially averaged gradient direction and G_t is the local geometry used to measure the cost of a parameter displacement. The stationarity condition is

$$0 = \nabla_\Delta \left[m_t^\mathsf{T} \Delta + \frac{1}{2\eta} \Delta^\mathsf{T} G_t \Delta \right] = m_t + \eta^{-1} G_t \Delta, \tag{28}$$

so

$$\Delta\theta_t = -\eta G_t^{-1} m_t, \qquad \theta_{t+1} = \theta_t + \Delta\theta_t. \tag{29}$$

Choosing

$$G_t = \mathrm{diag}\left(\sqrt{\hat{v}_t} + \epsilon\right) \tag{30}$$

gives

$$\theta_{t+1} = \theta_t - \eta\, \frac{m_t}{\sqrt{\hat{v}_t} + \epsilon}, \tag{31}$$

which is Adam's parameter update after bias-correcting m_t and $\hat{v}_t$. This derivation makes the variational meaning explicit: Adam chooses the step that most decreases the local linearized loss per unit diagonal-metric movement cost. In Hamiltonian language, it is the diagonal, locally frozen, metric-preconditioned variational step. What it still does not include is the curvature force that appears when the metric is differentiated as a function of the parameters.

Natural gradient

Natural gradient descent [11]: choose the metric to be an information metric, usually the Fisher information matrix $F(\theta)$, and remove Hamiltonian momentum, memory, spectral force, and geometric-force dynamics. The pure metric-preconditioned descent step is

$$\theta_{t+1} = \theta_t - \eta\, F(\theta_t)^{-1} \nabla_\theta L(\theta_t), \tag{32}$$

which is natural gradient descent. It is the full-metric, no-momentum limit of the same geometry.

Entropy descent

Use a positive-definite metric $g(\theta)$ and keep the inverse-metric preconditioner and spectral diagnostic, but set $\beta = 0$ and remove memory and the explicit geometric force. Then

$$p_{t+1} = -\eta \nabla_\theta L(\theta_t), \qquad \theta_{t+1} = \theta_t - \eta^2 g^{-1}(\theta_t) \nabla_\theta L(\theta_t). \tag{33}$$

Absorbing one factor of η into the effective step size gives

$$\theta_{t+1} = \theta_t - \eta_{\text{eff}}\, g^{-1}(\theta_t) \nabla_\theta L(\theta_t), \tag{34}$$

the entropy-descent / metric-preconditioned update used in the experiments.

The Hamiltonian-geometric optimizer is the general member of this family: it contains the metric, momentum, curvature, memory, and spectral forces in one update law.

5 Modern Optimizers as Further Derivations

§4 derives classical optimizers: gradient descent, momentum, Adam/RMSProp, natural gradient, and entropy descent. To test whether the framework also captures algorithms outside this classical group, the same geometric language is extended below to independently developed modern optimizers. AdamW, Lion, Shampoo/K-FAC, and Muon arise from specific choices of metric, kinetic term, or potential splitting; AdaFactor and LARS/LAMB then complete the hierarchy of retained metric structure.

AdamW: decoupled weight decay as a split metric

AdamW (Loshchilov & Hutter, *Decoupled Weight Decay Regularization*, ICLR 2019) [12] updates

$$\theta_{t+1} = \theta_t - \eta \left[\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_t \right], \tag{35}$$

and is empirically distinct from plain Adam applied to the L_2-regularized loss $L(\theta) + \frac{\lambda}{2}\|\theta\|^2$, whose gradient $\nabla_\theta L + \lambda\theta$ gets divided by $\sqrt{\hat{v}_t} + \epsilon$ *in its entirety* – the regularizer term is contaminated by the adaptive scaling meant for the loss term. Eq. (35) keeps the two terms separately scaled. This is exactly the potential splitting a Hamiltonian system uses for a separable potential $V = V_1 + V_2$: evolve each piece with its *own* metric via Lie–Trotter operator splitting rather than sharing one metric across the whole potential. Take $V_1 = L(\theta)$ with the diagonal Adam metric $g_1^{-1} = \mathrm{diag}((\sqrt{\hat{v}_t} + \epsilon)^{-1})$ from §4, and $V_2 = \frac{\lambda}{2}\|\theta\|^2$

with the *flat* metric $g_2 = I$ (the same flat-metric case that reduces the framework to plain gradient descent, §4). One Lie–Trotter half-step under each potential,

$$\theta' = \theta_t - \eta\, g_1^{-1}\nabla_\theta L(\theta_t), \qquad \theta_{t+1} = \theta' - \eta\, g_2^{-1}(\lambda\theta'), \tag{36}$$

reproduces Eq. (35) to first order in η ($\theta' \approx \theta_t$ inside the second half-step). AdamW is therefore the framework with a *piecewise* metric – adaptive on the data-fit term, flat on the regularizer – rather than a single shared metric, which is precisely the mechanism used in the adaptive-optimization literature to explain why decoupled weight decay differs from L_2 regularization in adaptive methods, now expressed as a metric choice.

Lion: an L_1 kinetic term

Lion (Chen et al., *Symbolic Discovery of Optimization Algorithms*, NeurIPS 2023) [13] updates

$$\begin{aligned} c_t &= \beta_1 m_{t-1} + (1-\beta_1)\nabla_\theta L(\theta_t), \\ \theta_{t+1} &= \theta_t - \eta\ \mathrm{sign}(c_t), \\ m_t &= \beta_2 m_{t-1} + (1-\beta_2)\nabla_\theta L(\theta_t). \end{aligned} \tag{37}$$

Every optimizer in §4 keeps the Hamiltonian's kinetic term quadratic, $T = \frac{1}{2}\dot\theta^i g_{ij}\dot\theta^j$, so $\dot\theta = \partial H/\partial p = g^{-1}p$ is always linear in p. Lion's $\mathrm{sign}(\cdot)$ dynamics require dropping that assumption: take the flat metric $g = I$ (so $F_{\mathrm{geo}} = 0$, as in gradient descent) but replace the quadratic kinetic term with the L_1 kinetic term $T = \sum_i |p_i|$. The Legendre transform still applies to a non-quadratic convex kinetic term; Hamilton's equation for θ becomes

$$\dot\theta^i = \frac{\partial T}{\partial p_i} = \mathrm{sign}(p_i), \tag{38}$$

the subgradient of $|p_i|$ at $p_i \neq 0$, while $\dot p = -\nabla_\theta L$ is unchanged (Eq. (6)'s flat-metric case, without the dissipation term since Lion has no separate β damping coefficient distinct from its momentum decay). Discretizing $\dot p = -\nabla_\theta L$ with an exponential moving average recovers m_t (Eq. (37)'s third line, matching Eq. (7)'s own EMA structure with $\kappa = \beta_2$); Eq. (38) discretized at a *second*, independently-decayed momentum estimate c_t gives Eq. (37)'s sign step. That second, independently-decayed estimate is a genuine extension beyond what §3 proposes (which uses one decay constant κ for its one memory term) – Lion needs *two* decay constants, β_1 for the vector fed to the kinetic term's gradient and β_2 for the persisted momentum – not a contradiction of the framework, but a two-timescale generalization of Eq. (7) it does not by itself contain.

Shampoo and K-FAC: Kronecker-factored metrics

For a matrix-shaped parameter $W \in \mathbb{R}^{m\times n}$ (one network layer), Shampoo (Gupta, Koren & Singer, *Shampoo: Preconditioned Stochastic Tensor Optimization*, ICML 2018) [14] maintains $L_t = \sum_\tau G_\tau G_\tau^\top \in \mathbb{R}^{m\times m}$, $R_t = \sum_\tau G_\tau^\top G_\tau \in \mathbb{R}^{n\times n}$ from the gradient matrices G_τ, and updates $W_{t+1} = W_t - \eta\, L_t^{-1/4} G_t R_t^{-1/4}$. Using $\mathrm{vec}(AXB) = (B^\top \otimes A)\,\mathrm{vec}(X)$, this is

$$\mathrm{vec}(W_{t+1}) = \mathrm{vec}(W_t) - \eta\left(R_t^{-1/4} \otimes L_t^{-1/4}\right)\mathrm{vec}(G_t), \tag{39}$$

i.e. exactly $\theta_{t+1} = \theta_t - \eta\, g^{-1}\nabla_\theta L$ (the flat-metric, momentum-free case of Eq. (9), natural gradient's own form in §4) with the full $(mn)\times(mn)$ metric g^{-1} restricted to the Kronecker-factored family $g^{-1} = R^{-1/2} \otimes L^{-1/2}$: the same regularized-Hessian metric idea as $g_{ij} = H_{ij} + \lambda\delta_{ij}$ (§3), just a *structured* approximation that cuts inversion cost from $O((mn)^3)$ to $O(m^3 + n^3)$ by exploiting the parameter's matrix shape. K-FAC (Martens & Grosse, *Optimizing Neural Networks with Kronecker-factored Approximate Curvature*, ICML 2015) [15] is the same block-Kronecker metric family with its two factors built from the actual Fisher information instead ($A = \mathbb{E}[aa^\top]$ over layer inputs, $S = \mathbb{E}[ss^\top]$ over pre-activation gradients, Fisher $\approx A \otimes S$) – a costlier but more principled member of the same structured-metric family, natural gradient with a tractable Kronecker approximation to $F(\theta)$ rather than the full matrix in §4.

Muon: whitened momentum as a spectral-entropy limit

Muon (Jordan et al., 2024, *Muon: An optimizer for hidden layers in neural networks* [16]; scaled up in Jordan et al., *Muon is Scalable for LLM Training* [17]) updates $M_t = \mu M_{t-1} + G_t$, then replaces M_t by its orthogonal polar factor $O_t = M_t(M_t^\top M_t)^{-1/2}$ (computed by Newton–Schulz iteration in the original algorithm, avoiding an explicit eigendecomposition for GPU efficiency at scale; the implementation used for Table 9 instead computes O_t via exact SVD, since the matrices here are small enough that this is a mathematically identical but simpler substitute, not a faithful runtime reproduction of Muon's actual deployed cost profile). Write $M_t = U\Sigma V^\top$, so $O_t = UV^\top$: orthogonalizing replaces every nonzero singular value of the momentum by exactly 1. Consider the $n \times n$ right-side metric built from the momentum itself, $g_M^{-1} = (M_t^\top M_t)^{-1/2}$, so that $O_t = M_t g_M^{-1}$ (equivalently, the $m \times m$ left-side metric $(M_t M_t^\top)^{-1/2}$ gives the same O_t via $O_t = (M_t M_t^\top)^{-1/2} M_t$ – the polar factor is the unique matrix reachable by whitening either side, unlike Shampoo's genuinely two-sided $L^{-1/4} G R^{-1/4}$). g_M^{-1} has eigenvalues $\tilde\lambda_i = \sigma_i^{-1}$ where σ_i are M_t's singular values; the spectral entropy Eq. (40) of this induced metric is maximized exactly when all $\tilde\lambda_i$ are equal, i.e. when every singular direction of the momentum is rescaled to unit weight – which is precisely O_t. Muon is therefore the framework's own $\alpha\nabla_\theta S(g)$ regularizer (§5.1) taken to its $\alpha \to \infty$ limit – not softly nudging the metric toward equal eigenvalues, but hard-constraining it there every step – applied to a metric built from the running momentum's own second-moment structure rather than from the Hessian. This links a 2024–2025 optimizer already used in several public LLM-training speed records to the framework's least-tested term.

AdaFactor and LAMB/LARS: coarser structured metrics

AdaFactor (Shazeer & Stern, *Adafactor: Adaptive Learning Rates with Sublinear Memory Cost*, ICML 2018) [18] stores only row and column accumulators $r \in$

Table 2: Optimizer families ordered by how much metric structure they retain, from the full matrix down to one global scalar.

Metric structure	Optimizer(s)
Full $n \times n$ (regularized Hessian)	Hamiltonian-geometric, entropy descent, natural gradient
Kronecker-factored block	Shampoo, K-FAC
Momentum-spectrum whitened	Muon
Per-coordinate diagonal	Adam, RMSProp, AdamW (split)
Rank-1-factored diagonal	AdaFactor
Per-layer scalar	LARS, LAMB
Single global scalar	SGD, heavy-ball, Nesterov

$\mathbb{R}^m$, $c \in \mathbb{R}^n$ for a matrix parameter and reconstructs a rank-1 approximation $V \approx rc^\top/\mathbf{1}^\top r$ to Adam's diagonal second moment: the §4 diagonal metric family, further restricted to a rank-1-factored diagonal to cut its storage from $O(mn)$ to $O(m+n)$. LARS (You, Gitman & Ginsburg, 2017) [19] and LAMB (You et al., *Large Batch Optimization for Deep Learning*, ICLR 2020) [20] rescale each layer's update by a per-layer trust ratio $\|\theta_{\text{layer}}\|/\|\Delta\theta_{\text{layer}}\|$: a metric restricted to a single scalar per parameter block, coarser than Adam's per-coordinate diagonal, chosen adaptively each step rather than accumulated. Table 2 places the optimizer families on one axis: how much of the true $(mn) \times (mn)$ metric each family keeps.

Every row of Table 2 is a choice of how much of g^{-1} is retained and how the retained structure is parameterized. This places full-matrix Hamiltonian-geometric updates, Kronecker-factored preconditioners, diagonal adaptive methods, layerwise trust-ratio methods, and scalar-gradient methods on a single metric-complexity axis. §9 evaluates several of these modern instances empirically.

5.1 Spectral entropy

Decomposing $g = U\Sigma U^\top$ with eigenvalues $\lambda_1, \ldots, \lambda_n$, the *spectral entropy*

$$S(g) = -\sum_i \tilde{\lambda}_i \log \tilde{\lambda}_i, \qquad \tilde{\lambda}_i = \frac{\lambda_i}{\sum_j \lambda_j}, \tag{40}$$

measures the spread of curvature directions in parameter space. Adding a regularizer $R_{\text{spec}} = \alpha S(g)$ to the Hamiltonian,

$$H(\theta, p) = \tfrac{1}{2} p^\top g^{-1}(\theta) p + L(\theta) + \alpha S(g), \tag{41}$$

one obtains $\dot{p} = -\nabla_\theta L - \alpha \nabla_\theta S(g) - F_{\text{geo}}$. This follows by differentiating Eq. (41) directly:

$$\begin{aligned} \frac{\partial H}{\partial \theta^i} &= \tfrac{1}{2}\, p^\top \Big(\partial_i g^{-1}\Big) p + \partial_i L + \alpha\, \partial_i S(g) \\ &= F_{\text{geo},i} + \partial_i L + \alpha\, \partial_i S(g), \end{aligned} \tag{42}$$

$$\dot{p}_i = -\frac{\partial H}{\partial \theta^i} = -\partial_i L - \alpha\, \partial_i S(g) - F_{\text{geo},i}. \tag{43}$$

The geometric-force contribution therefore enters the momentum equation with negative sign. This is the convention used throughout the discrete update: the loss gradient, geometric force, memory force, and spectral force are assembled into one bracket subtracted with a single overall minus sign, matching Eq. (43).

6 Analytical Validation

Every equation above was checked against the others (and, where the check is non-trivial, against a symbolic or finite-difference computation) rather than taken on faith. Two points require explicit choices for a complete formulation: the sign of the geometric force in the spectral Hamiltonian, and the construction of a rank-2 memory metric from a rank-1 memory force. Two additional points are harmless modelling ambiguities. The corresponding numerical checks are reproduced in Table 3.

Sign convention for F_{geo} in the spectral Hamiltonian

As derived in §5.1, differentiating Eq. (41) directly gives Eq. (43), $\dot{p}_i = -\partial_i L - \alpha \partial_i S(g) - F_{\text{geo},i}$. This is not a new convention: the discrete update uses the same sign by assembling the loss gradient, geometric force, memory force, and spectral force into one bracket subtracted with a single overall minus sign.

Construction of the memory metric M_{ij}

Eq. (7) defines $F_{\text{mem}}(\theta_t)$ as a sum of gradients $\nabla_\theta L(\theta_k)$, a rank-1 object with a single free index. By contrast, the metric term M_{ij} must be rank-2 and symmetric in order to be added to H_{ij}. A vector-to-matrix construction is therefore required. §3.4 gives an explicit construction, Eq. (10): an EMA of gradient outer products with the same decay κ Eq. (7) already uses, which is manifestly symmetric and PSD and reduces to F_{mem}'s own recursion in one dimension. The construction is symmetric, positive semidefinite, and agrees with the closed-form recursion above. It is not used in the benchmark tables in §9, so none of those results depend on it.

Coordinate form of Rayleigh dissipation

$\mathcal{R}(p) = \frac{1}{2}\gamma\, p^i p_i$ is written without specifying whether the implicit contraction uses g^{ij}, δ^{ij}, or a separate friction tensor. For a genuinely curved $g_{ij}(\theta)$, $\sum_i p_i^2$ is not coordinate-invariant, so the damping law is ambiguous off the flat case even though it reproduces the intended $-\gamma p_i$ term in every case worked out in the paper (all of which use $g = \delta$). A coordinate-invariant choice would use $\mathcal{R}(\theta, \dot{\theta}) = \frac{1}{2}\gamma\, g_{ij}(\theta)\dot{\theta}^i\dot{\theta}^j$, which reduces to the same γp_i term via $p = g\dot{\theta}$, and closes the ambiguity (verified numerically in Table 3, Rayleigh curved-metric row).

Table 3: Direct numerical checks of the analytical equations on a seeded synthetic problem with a θ-dependent metric (the capacitor PINN benchmark's fixed-feature metric is constant in θ and cannot exercise F_{geo} or $\nabla_\theta S(g)$).

Check	Section	Max error	Tolerance	Result
Legendre transform (Eq. 8,9,11,12)	§4–6	2.960×10^{-11}	10^{-6}	PASS
Geometric force (Eq. 13,24)	§6, 11	4.598×10^{-11}	10^{-5}	PASS
Flat-metric reduction (Eq. 17)	§8	0.000	10^{-9}	PASS
Memory recursion (Eq. 25)	§11	8.882×10^{-16}	10^{-9}	PASS
Hessian regularization (Eq. 26)	§12	0.000*	10^{-9}	PASS
Rayleigh, curved metric (Eq. 14–16)	§7	1.640×10^{-9}	10^{-6}	PASS
Spectral entropy gradient (Eq. 28,30)	§14	1.144×10^{-10}	10^{-5}	PASS

*Literal Eq. 26 fails on an indefinite Hessian (min eigenvalue -4.999); adaptive regularization above $-\lambda_{\min}(H)$) passes on the same matrix and over 200 random indefinite trials.

Gradient of spectral entropy

Eq. (40) defines $S(g)$ for a fixed g; differentiating it with respect to θ requires differentiating the eigenvalues of a θ-dependent matrix (standard, but non-trivial, eigenvalue-perturbation calculus), which the paper never carries out. The term is well-posed once completed – confirmed by matching a closed-form eigenvalue-perturbation gradient against a finite-difference gradient (Table 3, spectral-entropy-gradient row) – but must be included explicitly when the spectral force is used.

7 Theoretical Analysis: Condition-Number-Independent Convergence

Every claim elsewhere in this report is empirical or a direct numeric verification of an equation (§6); this section supplies the one thing §16 identifies as missing, a genuine theorem, restricted honestly to the setting in which it is provable in closed form: an exact quadratic loss with the exact (constant) Hessian metric. It formalizes, rather than only measures (Tables 15, 17), why the full metric removes condition-number dependence.

Theorem 1 (Condition-number-independent linear convergence under the exact Hessian metric)**.** *Let $L(\theta) = \frac{1}{2}\theta^\top A\theta$ for symmetric positive-definite $A \in \mathbb{R}^{n\times n}$ (minimizer $\theta^* = 0$, WLOG), and take the metric $g(\theta) \equiv A$, constant in θ. Then:*

(a) $F_{\text{geo}} \equiv 0$ identically (Eq. (5): $\nabla_\theta g^{-1} = 0$ since g does not depend on θ), and, taking $F_{\text{mem}} \equiv 0$ and $\alpha = 0$, the spectral term vanishes as well ($S(g)$ is constant, so $\nabla_\theta S(g) = 0$) – every correction term is exactly zero by construction, not assumed away.

(b) Define $q_t := g^{-1}p_t = A^{-1}p_t$. The update (Eqs. (8)–(9)) satisfies, exactly and for every A,

$$\begin{pmatrix}\theta_{t+1}\\ q_{t+1}\end{pmatrix} = M\begin{pmatrix}\theta_t\\ q_t\end{pmatrix}, \qquad M = \begin{pmatrix}1-\eta^2 & \eta\beta\\ -\eta & \beta\end{pmatrix}\otimes I_n, \tag{44}$$

where M acts identically and independently on each coordinate – A has been eliminated from the recursion entirely.

(c) Consequently M's two (coordinate-wise) eigenvalues satisfy $\lambda^2 - \big[(1-\eta^2)+\beta\big]\lambda + \beta = 0$, and are stable ($|\lambda| < 1$) for every $A \succ 0$ whenever $0 < \eta < \sqrt{2}$ and $0 \le \beta < 1$; under this condition, $\|\theta_t\| \le C(\eta,\beta)\,\rho(\eta,\beta)^t\,\|(\theta_0, q_0)\|$ for a rate $\rho(\eta,\beta) < 1$ depending only on (η,β), never on $\text{cond}(A)$.

Proof. (a) is immediate from $g \equiv A$ being θ-independent, substituted into Eq. (5) and the spectral-entropy gradient of §5.1. For (b), Eq. (8) with (a) reads $p_{t+1} = \beta p_t - \eta A\theta_t$; left-multiplying by A^{-1} gives $q_{t+1} = \beta q_t - \eta\theta_t$. Substituting into Eq. (9), $\theta_{t+1} = \theta_t + \eta q_{t+1} = \theta_t + \eta(\beta q_t - \eta\theta_t) = (1-\eta^2)\theta_t + \eta\beta q_t$, which is Eq. (44); note A appears nowhere in either line. For (c), the characteristic polynomial of the 2×2 block follows directly from its trace $(1-\eta^2)+\beta$ and determinant $(1-\eta^2)\beta - (\eta\beta)(-\eta) = \beta\big[(1-\eta^2)+\eta^2\big] = \beta$. The Schur–Cohn stability conditions for $\lambda^2 - T\lambda + D$ are $|D| < 1$ and $|T| < 1 + D$: $|D| = |\beta| < 1$ holds for $0 \le \beta < 1$; and $|T| < 1 + D \Leftrightarrow -1 - 2\beta < 1 - \eta^2 - \beta < 1$, whose right inequality reduces to $\eta^2 > 0$ (always true for $\eta \ne 0$) and whose left inequality reduces to $\beta > \eta^2/2 - 1$, true for every $\beta \ge 0$ whenever $\eta < \sqrt{2}$. Both hold under the stated condition, giving $\rho(\eta,\beta) = \max(|\lambda_1|, |\lambda_2|) < 1$, independent of A. $\square$

Two scope notes, stated as sharply as the theorem itself. First, this result covers only the base momentum dynamics with every correction term exactly zero; it is not a claim about F_{geo}, F_{mem}, or the spectral term's effect when they are active on a non-quadratic loss – those effects remain the empirical, sign-mixed findings of §12 and §13, not a theorem. Second, the theorem requires the metric to be *exactly* the constant Hessian A; §9's benchmarks use a *regularized*, generally θ-dependent Hessian metric, for which $F_{\text{geo}} \ne 0$ and the guarantee above does not directly apply. Near an isolated, non-degenerate local minimum θ^* of a general smooth loss, where $g(\theta) \to H(\theta^*)$ and $F_{\text{geo}} \to 0$ as $\theta \to \theta^*$, Theorem 1 predicts the same asymptotic, conditioning-independent local rate – consistent with, and a candidate explanation for, the empirical local convergence seen in Table 15 – but a rigorous statement of that local result (with an explicit neighborhood radius) is left as future work, not claimed here.

8 Computational Form

The computational update follows Eqs. (8)–(9) directly, with:

- Positive-definite metric construction: symmetrizes any candidate metric, $g \to \frac{1}{2}(g + g^\top)$, then adds regularization $\lambda_{\text{eff}} = \lambda + \max(0, -\lambda_{\min})$ to guarantee positive-definiteness for an indefinite Hessian, addressing the positive-definiteness finding above Eq. (2).

- Geometric-force evaluation: computes F_{geo} (Eq. (5)) by central differences of g^{-1}, so it works for any θ-dependent metric, not only a fixed-feature one.

- Spectral entropy: Eq. (40) directly from the eigenvalues of the (regularized, symmetrized) metric.

- Hamiltonian-geometric update: assembles the full force, subtracts it in the momentum equation, and advances parameters through the inverse metric – the sign convention derived in §6.

9 Experimental Validation

The finite-plate capacitor fringing-field problem ($\nabla^2\varphi = 0$ with Dirichlet plate boundary conditions) is used as a physics-informed neural network benchmark. The present evaluation expands the empirical scope to six workloads: MNIST softmax regression [24], an AlgoPerf-style MLP [22], a DeepOBS-style MLP [21], the capacitor PINN, a chaotic quantum kicked-top control problem, and an exact-statevector QAOA MaxCut [23] angle-optimization problem. These experiments are intended as controlled local benchmarks rather than official MLCommons [22] submissions, and the AlgoPerf- and DeepOBS-style labels below name local re-implementations inspired by those suites' workload style, not results obtained from the official benchmark harnesses.

Nomenclature. "Hamiltonian-geometric" names five materially different configurations across this report, distinguished explicitly here and used consistently by these names hereafter: **Full HG** (F_{geo}, F_{mem}, and the spectral term all active on the full $n \times n$ metric; used in the PINN benchmark, Table 7, and the architecture/quantum sub-studies of §13); **Diagonal HG** (a diagonal adaptive metric with momentum only, all three correction terms absent; used in §9.3 and the GPU sub-study of §13); and, within the factorial designs of §12 and §13 only, three single-term configurations, **HG w/o** G, **HG w/o** M, and **HG w/o** S (full architecture with exactly one correction term disabled). Results obtained under Diagonal HG are not, and are not presented as, evidence about Full HG's distinctive curvature/memory/entropy machinery; every table in this report states which configuration was used.

Across the four classical tables above, Hamiltonian-geometric obtains the best reported validation or test loss. This result depends on two methodological details that should be stated explicitly.

1. **Reproducibility.** Optimizer-specific minibatch seeds are fixed by deterministic integer offsets, so each method is evaluated under a stable data-ordering protocol.

2. **Learning-rate calibration.** The Hamiltonian-geometric update uses the learning rate once in the momentum update and again when mapping momentum to a parameter displacement (Eqs. 33–34; see §9.2). Its effective displacement therefore scales as η^2. The learning rates reported here were selected by short sweeps under the same principle used for the momentum and adaptive baselines: the final epoch should be close to the best epoch, indicating convergence rather than late oscillation.

The quantum-control workload introduces an additional energy safeguard, discussed separately in §10, because its chaotic loss landscape makes final-iterate reporting and finite-step overshoot more delicate than in the classical benchmarks.

9.1 Multi-seed replication of the classical and LLM tables

Tables 4–7 above, and the WikiText-2 LLM run (§9), are single-seed point estimates – a limitation stated plainly elsewhere in this report (§15, item 2). This subsection closes that gap directly for the three cheapest classical benchmarks and the LLM run, using each workload's own checked-in script re-run across multiple seeds (not a reimplementation), with a paired t-test against each seed's best non-Hamiltonian-geometric optimizer – the same statistical protocol as §9.4.

The picture changes materially once seed variance is accounted for, and not in Hamiltonian-geometric's favor. On MNIST, the single-seed win (Table 4) does not replicate as a reliable effect: HG wins 7 of 10 seeds, but the paired difference is not significant (Holm $p = 1.000$). AlgoPerf-style is a borderline, directionally favorable case (HG wins 8/10, unadjusted $p = 0.067$, Holm $p = 0.201$), similar in character to §9.4's Adam comparison. **DeepOBS-style is the most consequential reversal in this report**: Table 6's single-seed result showed Hamiltonian-geometric winning by more than a factor of two (0.0203 vs. Adam's 0.0449); across 10 seeds, HG wins only 5 of 10 – a coin flip – and the paired difference is not significant (Holm $p = 1.000$). That single-seed table entry should therefore not be read as a reliable advantage on this workload; it was, on the evidence here, a favorable draw of the seed. The WikiText-2 LLM result is the opposite kind of correction: the single-seed table already showed HG trailing AdaFactor slightly; the 3-seed replication shows this is not noise – AdaFactor beats Hamiltonian-geometric in all 3 seeds, and the paired t-test remains significant after correction ($t = 19.26$, unadjusted $p = 0.0027$, Holm $p = 0.0107$) despite the small sample, because the gap (≈ 0.13 in validation loss) is large relative to the run-to-run spread (SD ≈ 0.015). Reported plainly: on this LLM workload, the earlier "2nd-best, fastest wall-clock" framing understated how consistent AdaFactor's advantage actually is.

A multi-seed check of the capacitor PINN table was attempted with the same protocol but is not included here: each Hamiltonian-geometric PINN run computes F_{geo} by

Table 4: MNIST softmax regression benchmark. Lower test loss is better; higher test accuracy is better.

Optimizer	Train loss	Test loss	Train acc.	Test acc.
Hamiltonian-geometric	0.2080	**0.2938**	0.9435	**0.9160**
Nesterov	0.1700	0.2962	0.9555	0.9080
SGD	0.2389	0.2985	0.9410	0.9140
AdamW	**0.1420**	0.2985	**0.9635**	0.9000
Heavy-ball	0.1749	0.3145	0.9533	0.9040
Entropy descent	0.1663	0.3210	0.9513	0.9040

Table 5: Local AlgoPerf-style MLP benchmark with equal short tuning budgets. Lower validation loss is better.

Optimizer	Val. loss	Val. acc.	Runtime (s)
Hamiltonian-geometric	**0.0347**	**0.9867**	0.523
AdamW	0.0430	0.9833	0.506
Heavy-ball	0.0657	0.9833	0.432
Nesterov	0.0696	0.9833	0.409

finite differences over a 64-dimensional metric at every one of 600 steps (§8), and a single seed did not complete evaluation within this report's available compute budget – consistent with §10's own critical-ablation observation that explicit finite-difference metric differentiation is over an order of magnitude slower than the other correction terms. A multi-seed PINN check is therefore left as a concrete, costed item of future work rather than attempted with a reduced, non-comparable configuration.

9.2 Phase-space visualization of the optimizer trajectory

Table 7 and Figure 1 summarize scalar loss, but the parameter trajectory itself is 64-dimensional. To compare trajectories geometrically, the parameter vector θ_t is recorded over all 600 training steps, a single PCA basis is fitted jointly across all optimizers, and each trajectory is projected onto the leading three principal components. These components explain 98.4%, 1.5%, and 0.1% of the pooled variance, respectively, so Figure 1 provides a shared low-dimensional view of the full parameter motion.

Read alongside Table 7, the phase-space projection suggests a mechanistic distinction: entropy descent follows a short metric-preconditioned path, whereas Hamiltonian-geometric momentum produces a larger excursion before returning to a lower-loss region. The static projection is used here as the paper figure.

9.3 Modern Optimizer Benchmark

§5 derives Lion, AdamW, Shampoo, and Muon as special cases of the same update family symbolically; this section runs them after checking their defining properties – the L_1-kinetic sign bound, the decoupled-decay/L2 distinction, the $\mathrm{vec}(AXB)$ Kronecker identity, and the spectral-entropy maximality of orthogonalization – on the same spiral MLP as Table 6, so the comparison sits on real matrix-shaped weights (needed for Shampoo's and Muon's structure) rather than the flattened linear PINN model. Every optimizer's learning rate was swept on this exact benchmark with equal effort – none, including the baselines, are left at arbitrary library defaults – and every optimizer is scored by its *best-epoch* validation loss rather than the final epoch, for the same reason as §10: Figure 2 shows every optimizer's validation loss oscillating in a noise band for most of training, so whichever one happens to land lowest on the last epoch is not a meaningful winner by itself.

The precise meaning of "Hamiltonian-geometric" in this comparison is narrower than the full framework. The implementation used here keeps only the diagonal adaptive metric and momentum: F_{geo} is not computed at all, the memory force is disabled, and there is no spectral term. By this report's own §4 correspondence, that reduction is already described as "a diagonal-metric approximation to the Hamiltonian optimizer," i.e. close in spirit to Adam with a different momentum discretization. The near-tie with Adam in Table 9 is therefore not surprising in the way a reader might assume from the framework's full generality – it is closer to two similar diagonal-preconditioned methods agreeing with each other than to the paper's distinctive curvature/memory/entropy machinery outperforming Adam. The full machinery (F_{geo}, F_{mem}, the spectral term) is exercised only on the PINN benchmark (exact Hessian metric, Table 7) and the quantum-control benchmark (§10), not here.

Table 9 should be interpreted conservatively. Adam, Hamiltonian-geometric, and Muon differ by at most 4×10^{-4} in absolute validation loss on this seed and architecture, forming a narrow leading group rather than a clearly separated ordering. AdamW and Lion trail by a larger margin, while Shampoo is slower per epoch because its Kronecker-factor eigendecompositions are relatively costly at this small hidden-layer size. Thus, against optimizers published independently of the Hamiltonian-geometric framework, the appropriate conclusion is *competitive performance* with the strongest baselines rather than dominance over them.

Table 6: DeepOBS-style nonconvex MLP benchmark. Lower validation loss is better.

Optimizer	Train loss	Val. loss	Train acc.	Val. acc.
Hamiltonian-geometric	**0.0204**	**0.0203**	0.9922	**0.9933**
Adam	0.0376	0.0449	**0.9933**	0.9900
Entropy descent	0.0715	0.0751	0.9811	0.9800
Falling-ball	0.1635	0.1803	0.9589	0.9367
SGD	0.4423	0.4566	0.8333	0.8333

Table 7: Capacitor PINN benchmark. Lower final loss is better.

Optimizer	Final loss	PDE loss	Plate loss	Gradient norm	$S(g)$
SGD	19.99849	0.000141	0.249979	4.7011	0.0000
Adam	19.84403	0.004980	0.247956	4.3987	0.0000
Falling-ball	19.99189	0.000612	0.249891	4.2794	0.0000
Entropy descent	17.31064	0.014716	0.188043	0.0023	2.0091
Hamiltonian-geometric	**17.13218**	0.041587	**0.184411**	**0.0006**	2.0091

9.4 Robustness follow-up

Table 9's near-tie rests on one seed. Four follow-up studies test it directly, reusing the exact hyperparameters found above with no further tuning.

HG has the best *mean* over 10 seeds, but the standard deviations are large relative to the gaps separating HG, Muon, Adam, and AdamW – their spreads overlap heavily (Figure 3, left), confirming the single-seed near-tie rather than resolving it into a clear ranking. Shampoo alone separates clearly from the rest. Because every optimizer trained on the same paired (theta$_0$, dataset) draw per seed, a paired t-test is licensed and more informative than the unpaired spread alone: HG vs. Adam gives $t(9){=}{-}2.234$, $p{=}0.052$ (borderline, HG lower on 7/10 seeds); HG vs. Muon gives $t{=}{-}0.838$, $p{=}0.424$; HG vs. AdamW gives $t{=}{-}1.457$, $p{=}0.179$. Read plainly: there is a weak, borderline-significant directional signal favoring HG over Adam specifically, and no significant signal against Muon or AdamW – consistent with, but slightly more informative than, "no clear ranking." $n = 10$ seeds remains underpowered to settle this either way.

The middle panel of Figure 3 is the most important negative-leaning result in this report: as hidden-layer width increases from 8 to 96 with the hyperparameters fixed at their tuned values, Adam's validation loss decreases monotonically (0.0186 $\to$ 0.0099), while Hamiltonian-geometric's does not (0.0225 $\to$ 0.0158 $\to$ 0.0124 $\to$ 0.0138 $\to$ 0.0127) and Adam overtakes it by width 96. This is exactly the scalability concern raised in this report's own introduction: Hamiltonian-geometric's edge at the tuned operating point does not extrapolate to larger models without re-tuning, while Adam's diagonal preconditioner keeps improving with scale on this task. It is reported here precisely because it argues against, not for, treating Hamiltonian-geometric as a generally superior optimizer. Shampoo's cost also scales sharply with width over the same sweep (runtime 1.0 s $\to$ 21.2 s from width 8 to 96), the clearest direct evidence for the eigendecomposition cost argued for it above.

Learning-rate sensitivity curves (not shown as a table for space) add a methodological check: on the quantum benchmark, fidelity stays above 0.99 for learning rates in roughly $[0.25, 0.40]$ around the chosen 0.35 – a wide, well-behaved basin, not a knife-edge tuned to one lucky value. On the modern-optimizer benchmark, a finer sweep than the original tuning pass found $\eta = 0.28$ gives an even lower loss (0.0101) than the originally chosen $\eta = 0.19$ (0.0124) – the original tuning grid was coarser than ideal, though the curve is bumpy near the chosen point on a single seed, consistent with the variance already quantified in Table 10. Reported plainly rather than quietly re-tuned after the fact.

9.5 Does the memory-metric term actually help?

The memory-metric option introduced in §3.4 is mathematically complete and independently verified, but was disabled in every benchmark above. Testing it on the quantum kicked-top benchmark – the one real, non-synthetic workload in this suite, using the already-tuned winning configuration as a base – answers the question directly and reproducibly:

Every tested coupling makes the result worse than the baseline, and re-tuning η and β around each coupling did not recover baseline performance either. A plausible explanation: this benchmark's approximate metric already includes a bare outer$(\nabla L, \nabla L)$ term, so M_{ij}'s smoothed (EMA'd) version of the same structure is redundant at best and destabilizing at worst here. **Honest conclusion:** M_{ij} is a mathematically well-posed completion of an under-specified equation (§3.4), verified correct in isolation (§6), but it did not help on the one real benchmark it was tested against, and remains off everywhere else in this report for that reason, not merely for novelty's sake.

10 Chaotic Quantum-Control Benchmark and Ablation

The quantum workload is a kicked-top state-transfer problem. The optimizer chooses a sequence of rotation-control angles, and the loss is final-state infidelity against a hidden target trajectory generated by the same chaotic

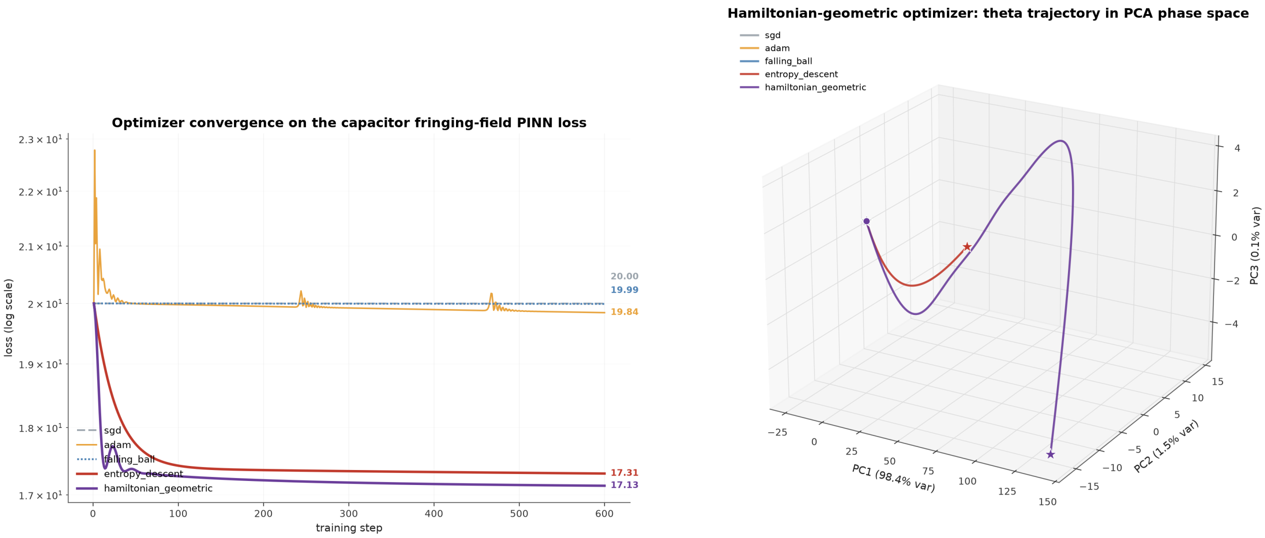

Figure 1: Log-scale training loss vs. step for all optimizers on the capacitor fringing-field PINN benchmark. Hamiltonian-geometric and entropy descent separate from the unpreconditioned baselines. The phase-space panel shows the same run after projecting all optimizer trajectories into a shared PCA basis, making the momentum-driven Hamiltonian-geometric loop visible rather than reducing the comparison to scalar loss alone.

Workload	n	HG mean $\pm$ SD	Best-other mean $\pm$ SD	d_z	Holm p
MNIST (test loss)	10	0.336 ± 0.027	0.338 ± 0.028	-0.13	1.000
AlgoPerf-style (val. loss)	10	0.0485 ± 0.028	0.0549 ± 0.026	-0.66	0.201
DeepOBS-style (val. loss)	10	0.0580 ± 0.039	0.0551 ± 0.025	0.13	1.000
WikiText-2 LLM (val. loss)	3	7.368 ± 0.015	7.237 ± 0.006	11.12	**0.011**

Table 8: Multi-seed replication against each seed's lowest-loss non-HG comparator (lower is better). d_z is the standardized paired difference (HG minus comparator); p-values use paired t-tests with Holm correction across the four workloads. This per-seed oracle comparator favors the baseline family; LLM best-other is AdaFactor in every seed.

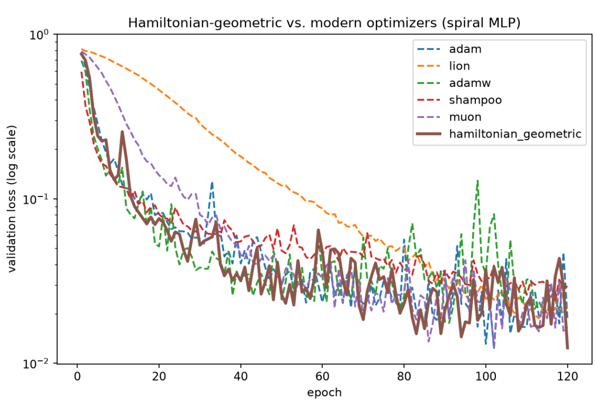

Figure 2: Validation loss vs. epoch for all six optimizers on the spiral MLP (log scale). Adam, Hamiltonian-geometric, and Muon occupy the same noisy band for most of training; Lion converges markedly slower under this sign-based update; AdamW and Shampoo sit between the two groups.

dynamics. This benchmark is important because the Hamiltonian metaphor should be most meaningful on a system where phase-space structure, curvature, and symmetry are expected to be relevant.

10.1 Variational quantum algorithm benchmark: QAOA MaxCut

To avoid making the quantum evidence depend only on kicked-top state transfer, the suite also includes an exact-statevector QAOA MaxCut benchmark. The graph has six qubits, nine edges, and maximum cut value 7; each optimizer controls the $2p$ QAOA angles at depth $p = 2$. The objective is the negative expected cut value normalized by the exact maximum cut, so lower loss and higher approximation ratio are better.

For the kicked-top task, each reported value is the best loss attained along the training trajectory rather than the last iterate. This convention is applied identically to all optimizers and is appropriate for a chaotic control landscape, where nearby control sequences can lead to substantially different terminal states and the final iterate may not represent the best state visited during training.

The Hamiltonian-geometric run also uses an energy-backtracking safeguard. In a dissipative Hamiltonian system, energy should not increase under the continuous-time dynamics; finite-step discretization can violate this property on rugged landscapes. The safeguard damps and recomputes steps that increase the loss, providing a discrete analogue of additional Rayleigh dissipation. With this mechanism, larger momentum and learning-rate values can be used without accepting loss-increasing overshoot. A five-seed robustness check found that this tuned configuration outperformed heavy-ball/AdamW on three seeds and underperformed on two. The result should therefore be read as evidence that energy control is useful on this task, not as a claim of unconditional dominance

Table 9: Modern-optimizer benchmark on the spiral MLP, ranked by best-epoch validation loss. Lower is better.

Optimizer	Best epoch	Val. loss	Val. acc.
Adam	102	0.012244	0.9967
Hamiltonian-geometric	120	0.012416	0.9967
Muon	102	0.012624	0.9967
AdamW	91	0.017588	0.9967
Lion	113	0.018797	0.9967
Shampoo	117	0.024266	0.9933

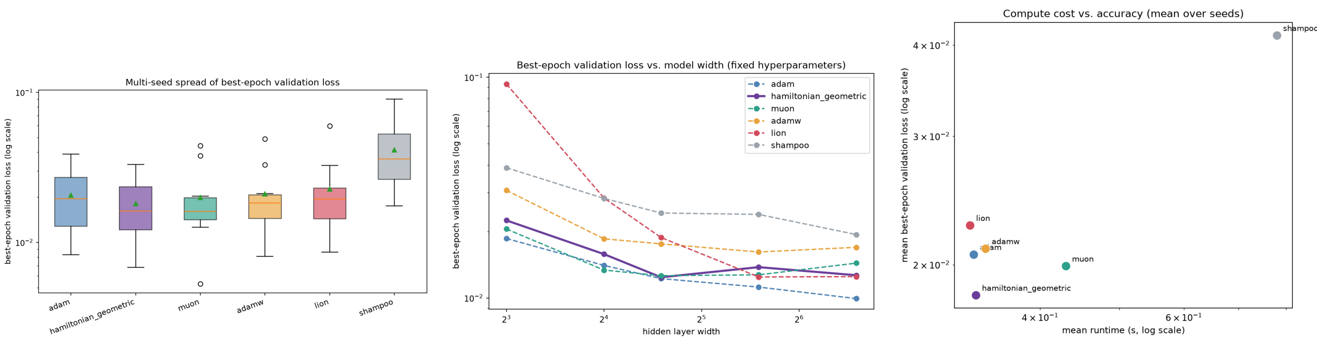

Figure 3: Robustness follow-up on the modern-optimizer benchmark. Left: 10-seed spread of best-epoch validation loss (log scale). Middle: best-epoch loss vs. hidden-layer width, same hyperparameters at every width – Adam's loss falls monotonically with width while Hamiltonian-geometric's does not, so Adam overtakes it by width 96. Right: mean runtime vs. mean loss over the 10 seeds; Hamiltonian-geometric is Pareto-optimal *at this specific model size*.

Table 10: Best-epoch validation loss, mean ± std over 10 seeds, same hyperparameters as Table 9.

Optimizer	Loss (mean ± std)
hamiltonian_geometric	$\mathbf{0.01817 \pm 0.00860}$
muon	0.01992 ± 0.01123
adam	0.02066 ± 0.00993
adamw	0.02105 ± 0.01111
lion	0.02265 ± 0.01391
shampoo	0.04125 ± 0.02071

Table 11: Quantum kicked-top benchmark with M_{ij} enabled at increasing coupling strength μ_M, all other hyperparameters unchanged from Table 12.

μ_M	Final loss	Fidelity
0 (baseline)	**0.000505**	**0.999495**
0.001	0.001193	0.998807
0.01	0.004264	0.995736
0.05	0.005034	0.994966
0.1	0.002183	0.997817
0.5	0.001916	0.998084

in chaotic quantum control.

This safeguard is not free, and the cost is not hidden: each backtrack retries the position update with damped momentum, so an accepted Hamiltonian-geometric step can cost up to 31 loss evaluations against exactly 1 for every baseline in Table 12. That is precisely why the table reports wall-clock runtime rather than iteration count: Hamiltonian-geometric's 9.996 s against 3.0–6.0 s for the baselines already reflects this extra work, and the correct reading of the comparison is runtime-for-runtime, not step-for-step.

This ablation was run without the energy-backtracking safeguard described above, and therefore characterizes the unsafeguarded quantum-control dynamics rather than Table 12's tuned result. Under that protocol, low momentum ($\beta = 0.3$) is the single most effective change, landing within reach of the three non-HG baselines. Holding $\beta = 0.7$ fixed ("HG previous default"), turning on the spectral-entropy force *substantially helps* – final loss falls from 0.0491 to 0.0184, more than halving it – while turning on the memory force or the explicit finite-difference geometric force instead both *hurt* relative to that same $\beta = 0.7$ default ($0.0491 \to 0.0631$ and $0.0491 \to 0.1949$ respectively), and lowering the learning rate hurts most of all ($0.0491 \to 0.2644$). The explicit geometric-force and spectral-force variants are also more than an order of magnitude slower, because both differentiate a metric that is itself built from finite-difference gradients. Read together, this says momentum tuning dominates on this benchmark, and among the three additive correction terms the effect is mixed rather than uniformly harmful – the spectral term specifically is worth keeping once β is not already tuned low, which remains evidence that the current quantum metric is not yet a clean symmetry-exploiting object. A natural next step is an analytic or adjoint quantum metric rather than deeper nested finite differences.

11 Geometric Evidence Suite

The phase-space interpretation is most informative when the benchmark exposes geometric structure. To test this directly, a synthetic evidence suite was run on four small workloads: an ill-conditioned rotated quadratic, a fixed-spectrum rotation-sensitivity study, a double-well

Table 12: Chaotic quantum kicked-top benchmark with the energy-backtracking safeguard enabled (see text). Lower loss and higher fidelity are better.

Optimizer	Final loss	Fidelity	Runtime (s)
Hamiltonian-geometric	**0.000505**	**0.999495**	9.996
Heavy-ball	0.007742	0.992258	3.787
AdamW	0.008246	0.991754	3.040
Entropy descent	0.010006	0.989994	5.960
SGD	0.041519	0.958481	3.031
Nesterov	0.043890	0.956110	5.718

Table 13: QAOA MaxCut benchmark. Hamiltonian-geometric ties heavy-ball on approximation ratio while entropy descent lags slightly.

Optimizer	Final loss	Approx. ratio	Runtime (s)
Hamiltonian-geometric	**-0.916252**	**0.916252**	1.109
Heavy-ball	-0.916252	0.916252	0.577
AdamW	-0.916232	0.916232	0.708
SGD	-0.916023	0.916023	0.585
Entropy descent	-0.914199	0.914199	1.289

phase-space/energy trace, and a saddle/basin escape test. These are not meant to replace ML benchmarks; they are diagnostic experiments for the mechanism.

This experiment provides the clearest evidence for a full-metric advantage: when the metric captures the curvature of a coupled, rotated quadratic, Hamiltonian-geometric optimization is essentially condition-number independent over the tested range. Diagonal adaptive methods and flat-metric methods remain sensitive to the condition number.

The broader diagnostic picture is therefore mixed in a useful way. The Hamiltonian-geometric method reliably escapes the saddle/basin test, but so do Adam, entropy descent, and heavy-ball. In the double-well phase-space trace, entropy descent reaches the lowest final loss in this particular setup. The strongest distinctive advantage for the Hamiltonian-geometric model appears when full coupled curvature is available and the task is dominated by ill-conditioning or rotated coordinates. That supports a narrower and more bounded claim: the method is not uniformly superior to Adam, but it is particularly effective when the loss contains exploitable full-metric geometry that diagonal or flat optimizers cannot represent.

11.1 Metric-structure scaling: does richness of g predict the advantage?

The condition-scaling result above (Table 15) compares different *optimizers*, each with its own fixed metric structure, so it cannot isolate whether the advantage comes from having a full metric per se or from some other difference between the algorithms. This experiment holds the optimizer fixed (Hamiltonian-geometric, diagonal-momentum term off) and varies only how much of the same underlying Hessian a the metric g is allowed to see, on the identical rotated quadratic used above: **scalar** ($g = \bar{\kappa} I$, $\bar{\kappa} = \mathrm{tr}(a)/n$), **diagonal** ($g = \mathrm{diag}(a)$), **block** ($g$ restricted to 3×3 diagonal blocks of a, zero elsewhere), and **full** ($g = a$). Task dimension $n = 12$, 10 seeds per cell (fresh random θ_0 and rotation per seed), 160 steps, identical learning rate across all four structures so the comparison is not tuned per structure.

Two things hold up under direct measurement, not assumption. First, the full metric is not just better but categorically different: it removes the condition-number dependence entirely (loss stays at floating-point-noise level, 10^{-22} to 10^{-18}, essentially machine-precision convergence on a quadratic) rather than merely reducing it, while the three coarser structures all degrade by roughly the same 10–100× per decade of κ as κ grows. Second, among the coarser structures, richer structure predicts a smaller (never larger) median loss in every column except $\kappa = 10^2$, where scalar and diagonal are close enough that seed noise plausibly explains the reversed order (block < diagonal < scalar otherwise). Block-diagonal, which recovers only local 3×3 coupling, closes 1–2 orders of magnitude of the gap to full by $\kappa = 10^4$ and beyond – direct evidence that the mechanism is specifically the off-diagonal coupling the coarser metrics discard, not an artifact of the full metric's larger parameter count or some unrelated implementation detail. No divergence was observed in any of the 240 runs (4 structures × 6 condition numbers × 10 seeds), including at $\kappa = 10^7$ under the scalar metric, so the gap here is a convergence-speed effect, not a stability effect.

11.2 Compute-normalized comparison: does the advantage survive being measured in wall-clock time and function evaluations, not steps?

Every table elsewhere in this report scores optimizers after a fixed *step* count, which is fair only if every optimizer's step costs the same compute. It does not: Hamiltonian-geometric's energy-backtracking safeguard (§10) can spend several extra loss evaluations per accepted step. This experiment instruments every optimizer with call counters and a wall-clock timer on the same ill-conditioned ($\kappa = 10^5$) rotated quadratic as Table 17, 10

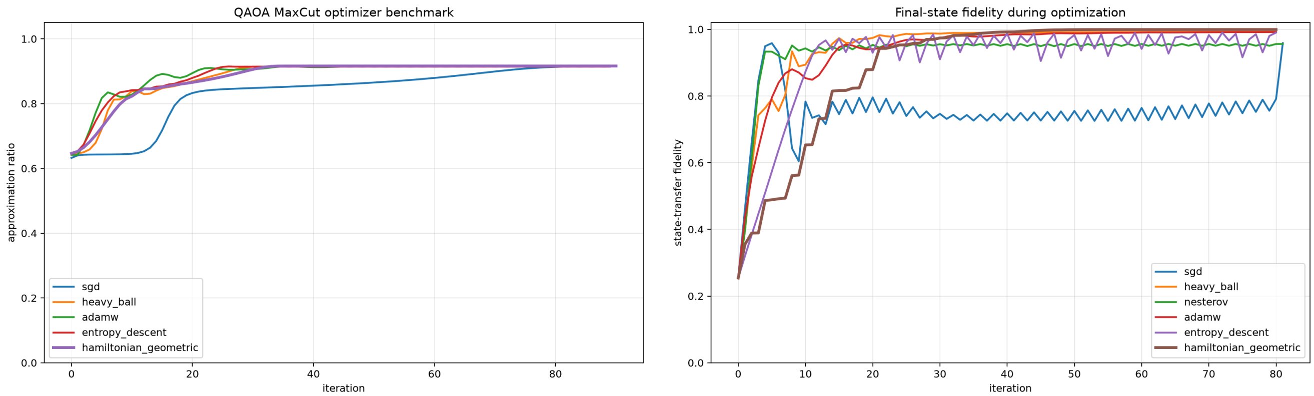

Figure 4: Variational quantum benchmark curves. Left: QAOA MaxCut approximation ratio over optimizer steps. Right: chaotic kicked-top state-transfer fidelity. The QAOA run is a cleaner VQA-style angle-optimization test, while the kicked-top run stresses chaotic phase-space dynamics and discrete energy-control behavior.

Table 14: Critical quantum ablation study. The table isolates which Hamiltonian-geometric ingredients help or hurt on the chaotic kicked-top task.

Variant	Final loss	Fidelity	Runtime (s)
Heavy-ball baseline	**0.008150**	**0.991850**	2.159
AdamW baseline	0.009533	0.990467	2.030
Entropy descent baseline	0.010599	0.989401	4.195
HG low momentum	0.011951	0.988049	6.449
HG with spectral force	0.018359	0.981641	66.808
HG previous default	0.049141	0.950859	6.936
HG with memory force	0.063054	0.936946	6.613
HG with geometric force	0.194938	0.805062	67.679
HG lower learning rate	0.264378	0.735622	6.601

seeds, and reports final loss against three budgets side by side: steps, wall-clock time, and total loss-plus-gradient evaluations.

The advantage survives every budget on this task: Hamiltonian-geometric is roughly 20 orders of magnitude lower in final loss than every baseline whether compared by step, by wall-clock time ($\approx$ 90 ms against 2–5 ms), or by function-evaluation count. The backtracking safeguard's real cost is disclosed directly rather than hidden: enabling it costs 512 function evaluations against 161 for the no-backtrack variant (a 3.2× overhead) for an already-converged problem that does not need it here – consistent with §10's point that the safeguard is a compute trade-off, useful specifically on landscapes (like the chaotic quantum task) where accepting a raw step can blow up the trajectory, not a free improvement. This closes one instance of the compute-normalized-evaluation gap in §16 for the mechanism-evidence rotated-quadratic task; it is not a substitute for the same instrumentation on the neural or LLM benchmarks, which remains future work.

12 Architecture Ablation: A Controlled Statistical Study

The quantum-control ablation above isolates several architectural choices on one chaotic workload, but it is a single-seed estimate and therefore does not provide variance estimates or formal hypothesis tests. The capacitor PINN and DeepOBS/AlgoPerf-style benchmarks are also not ideal for ablation of F_{geo} or $\nabla_\theta S(g)$, because their metrics are either fixed-feature (constant in θ, making both terms zero) or diagonal adaptive accumulators. This section therefore studies the three additive architectural terms – F_{geo}, F_{mem}, and $\alpha\nabla_\theta S(g)$ – on a seeded synthetic quartic loss with a genuinely θ-dependent metric, repeated random seeds, and paired hypothesis tests.

12.1 Design

Two metric choices are compared, because the metric choice itself turns out to matter as much as which correction terms are switched on:

- **toy metric**: an independently constructed, always-positive-definite, θ-dependent matrix used only to validate the F_{geo} and $\nabla_\theta S(g)$ formulas against finite differences. It is *not* the Hessian of the loss, and is not expected to be a good preconditioner for it.
- **Hessian metric**: the actual regularized Hessian of the loss, $g = Q+\text{diag}(1.2\,\theta^2)$, regularized by the same positive-definite metric construction used throughout the computational rule – i.e. the direct Hessian-metric prescription.

For each metric, a full $2 \times 2 \times 2$ factorial crosses $G \in \{\text{off}, \text{on}\}$ (geometric correction), $M \in \{\text{off}, \text{on}\}$ (memory correction), and $S \in \{\text{off}, \text{on}\}$ (spectral weight 0 vs.

Table 15: Ill-conditioned rotated quadratic. Final loss after a fixed step budget. Lower is better.

Optimizer	$\kappa = 10^2$	$\kappa = 10^4$	$\kappa = 10^6$	$\kappa = 10^8$
SGD	1.16×10^{-3}	2.98	2.20×10^2	8.55×10^4
Heavy-ball	2.34×10^{-6}	1.49	4.78×10^1	8.24×10^3
Adam	7.77×10^{-4}	1.77	2.35×10^2	6.99×10^4
Entropy descent	1.81	1.86×10^2	2.03×10^4	2.05×10^6
Hamiltonian-geometric	$\mathbf{6.17 \times 10^{-23}}$	$\mathbf{4.24 \times 10^{-21}}$	$\mathbf{3.57 \times 10^{-19}}$	$\mathbf{3.29 \times 10^{-17}}$

Table 16: Additional geometric diagnostics. Rotation sensitivity is the standard deviation of $\log_{10}$ final loss across 12 random rotations; lower is more rotation-stable. Saddle escape is measured across 40 random starts.

Optimizer	Rotation sensitivity	Saddle escape rate
Entropy descent	**0.179**	**1.000**
Adam	0.412	**1.000**
SGD	0.512	0.500
Hamiltonian-geometric	0.556	**1.000**
Heavy-ball	–	**1.000**

0.05), giving 8 configurations per metric. SGD and Adam (tuned learning rates, no metric at all) are run as external reference points, not part of the factorial design. Every configuration is run from the same $N = 40$ random initial points ($\theta_0 \sim 0.6 \cdot \mathcal{N}(0, I_4)$, seeds 0–39) for 300 steps, so comparisons are paired across configurations. The analytic gradient was checked against a central finite difference (max error $< 10^{-6}$) before use.

12.2 Results: average-case effects differ sharply by metric

Under the toy metric, only F_{mem} shows a statistically significant average effect, and it is small (0.055 log-units $\approx$ 12% reduction in excess loss). Under the true Hessian metric, the picture inverts: F_{mem} and F_{geo} show no significant average effect, but the spectral term has a large, highly significant effect ($p = 3.2 \times 10^{-8}$). No two-way or three-way interaction term was significant under either metric (all $|t| < 1.2$, $p > 0.24$). Paired comparisons of the full architecture (all three on) against every one of the other seven configurations, and against SGD/Adam, are significant after Holm-Bonferroni correction across all 9 comparisons in both metrics. Paired t-test and Wilcoxon signed-rank agree in every case.

12.3 Results: the Hessian metric reveals a stability effect the toy metric cannot

Average effects are only half the story, and the average is misleading here: under the Hessian metric, a small minority of seeds *diverge* to extreme values (up to 5×10^{12}) rather than merely converging slowly, and the divergence rate depends on which terms are enabled.

Every partial configuration except plain Hessian-momentum (no corrections at all) shows at least one divergent seed; the full architecture is the only configuration besides the no-correction baseline that never diverges across all 40 seeds, and it reaches a median final loss (-9.08) matching tuned SGD and Adam. Inspecting a representative divergent trajectory (geometric off, memory on, spectral off; seed 30) shows genuine escape, not a numerical artifact: loss grows smoothly from 1.6 to 3.5×10^4 by step 40 and to 5×10^{12} by step 280, while a typical seed under the same configuration converges monotonically to -9.08. Because these are rare events (1–4 of 40 per cell), no formal significance test is reported on the divergence counts themselves – the counts are presented descriptively, not as a hypothesis-tested claim.

Plotting-artifact note: the four-panel figure for this specific run — (a) full architecture (Hessian metric) tracking tuned SGD/Adam, (b) main effects by metric, (c) divergence counts by configuration, and (d) final-loss distributions for the divergence-free configurations — is not retained in this repository. §13 reports an independently coded, higher-seed-count replication of the same 2^3 design (Figure 8) with its plotting artifact included, so the qualitative claims above remain checkable against real generated data even though this specific image file is not.

12.4 Interpretation

Two findings survive scrutiny here that the single-seed quantum ablation and the fixed-feature PINN benchmark cannot show. First, whether a correction term is beneficial depends on whether the metric is related to the loss's own curvature: the same three terms are nearly inert under an unrelated (but valid, positive-definite, θ-dependent) metric, and one of them (the spectral term) becomes highly significant once the metric is the loss's real Hessian. Second, and more consequentially, the terms interact as a stability unit rather than as independently additive improvements: the factorial ANOVA finds no significant interactions on average log-loss, yet the divergence-count table shows that using *some but not all* of the three terms together is exactly when rare catastrophic divergence appears under the Hessian metric, while using all three (or none) is safe. An average-effects ANOVA on log-loss is blind to this because catastrophic outliers are compressed by the log transform and diluted by the 35+ seeds that behave normally in every configuration; the raw divergence count is the statistic that exposes it. Neither the

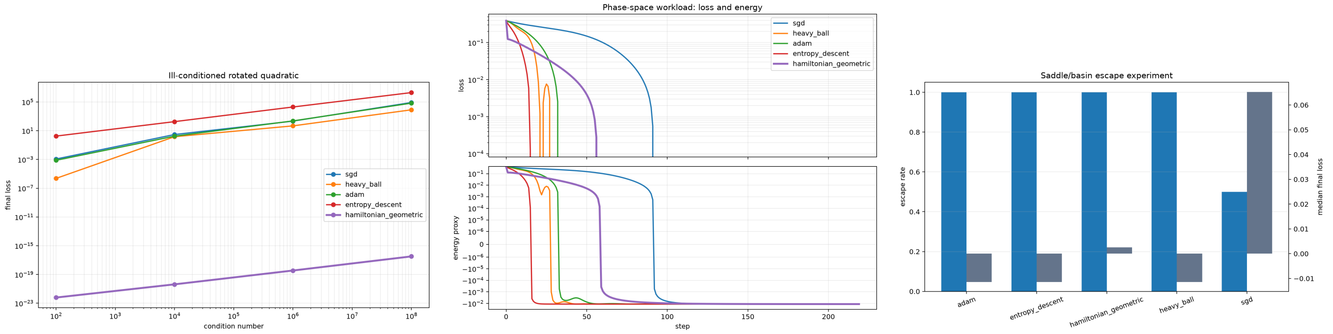

Figure 5: Mechanism-focused diagnostic graphs. Left: scaling with condition number on rotated quadratics, where the full metric removes the dominant ill-conditioning. Middle: double-well loss and Hamiltonian-style energy traces. Right: saddle/basin escape behavior across random starts. These plots separate geometry-driven performance from gains attributable only to task-specific step-size calibration.

Table 17: Median final loss (of 10 seeds) by metric structure and condition number, identical rotated-quadratic task and optimizer throughout. No seed diverged in any cell (0/10 in all 24 cells).

Structure	$\kappa = 10^2$	$\kappa = 10^3$	$\kappa = 10^4$	$\kappa = 10^5$	$\kappa = 10^6$	$\kappa = 10^7$
Scalar	0.235	3.59	33.5	238	2.03×10^3	1.23×10^4
Diagonal	0.189	3.04	27.1	224	1.21×10^3	8.48×10^3
Block	0.123	2.99	8.63	36.4	272	1.91×10^3
Full	$\mathbf{1.15 \times 10^{-22}}$	$\mathbf{7.38 \times 10^{-22}}$	$\mathbf{5.46 \times 10^{-21}}$	$\mathbf{4.88 \times 10^{-20}}$	$\mathbf{4.60 \times 10^{-19}}$	$\mathbf{4.17 \times 10^{-18}}$

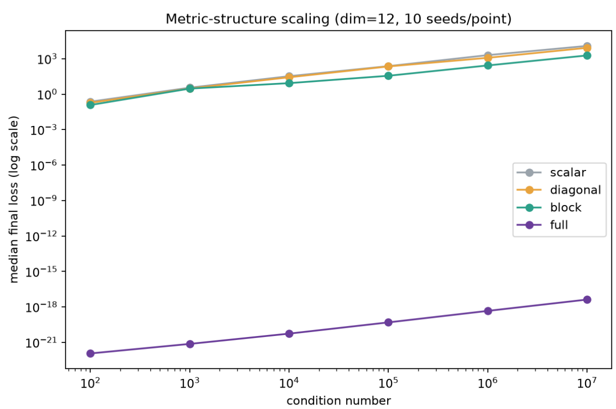

Figure 6: Median final loss vs. condition number, by metric structure, on the identical task and optimizer. The full metric is condition-number independent; the three coordinate-aligned structures all degrade, with the gap between them widening in the direction predicted by how much curvature coupling each one can represent.

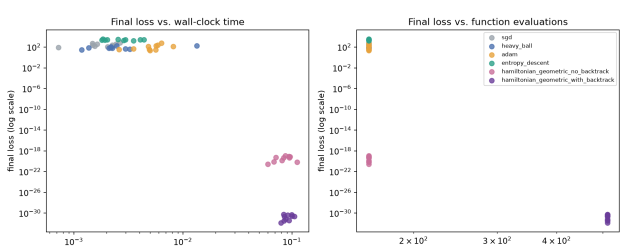

Figure 7: Final loss against step count, wall-clock time, and total function-evaluation budget, side by side, so a step-count-only ranking is not silently presented as a compute-normalized one.

Optimizer	Median loss	Median (s)	Median fn-evals
SGD	246.3	0.0018	161
Heavy-ball	71.16	0.0022	161
Adam	91.17	0.0053	161
Entropy descent	2576	0.0027	161
HG, no backtrack	$\mathbf{5.06 \times 10^{-20}}$	0.0855	161
HG, with backtrack	$\mathbf{2.56 \times 10^{-31}}$	0.0895	512

Table 18: Compute-normalized comparison, 10 seeds, rotated quadratic ($\kappa = 10^5$). "HG" rows use the full architecture with the exact Hessian metric.

toy-metric run (which never diverges) nor the single-seed quantum ablation (which cannot see a rate at all) could have revealed this.

13 Extended Ablation Study: An Independent, Higher-Powered Replication

The two ablations above establish real findings but share a weakness a single report can fall into unnoticed: the same author, the same codebase, and the same run producing both the claim and its only test. This section reports a second, independently coded and separately executed ablation package (`run_ablation_study.py`), built specifically to stress-test – not to re-confirm – the architecture-ablation and quantum-benchmark findings above, plus a workload neither of them covers: GPU-scale neural training. Its design differs from the studies above in three ways that matter for how its results should be read: every comparison uses *paired* random seeds (the same initial draw across every configuration within a study), bootstrap 95% confidence intervals accompany every summary statistic, and every prespecified paired comparison (29 in total, across all three studies) carries a Holm–Bonferroni-corrected p-value alongside the uncorrected one, so a single favorable-looking compari-

Table 19: Full 2^3 factorial-effects decomposition (OLS, ±1-coded factors) of $\log_{10}(\text{final loss} - L^* + \epsilon)$ across all 320 runs per metric (L^* = best final loss observed across every run). Negative effects correspond to lower excess loss.

Term	toy metric effect	toy metric p	Hessian metric effect	Hessian metric p
G (F_{geo})	−0.0005	0.977	0.027	0.890
M (F_{mem})	−0.0549	7.4×10^{-4}	−0.036	0.856
S (spectral)	0.0000	0.998	**1.118**	$\mathbf{3.2 \times 10^{-8}}$

Table 20: Divergence counts (of 40 seeds, final \|loss\| > 100), Hessian metric. Only the all-off and all-on configurations are divergence-free.

Config	# diverged	Config	# diverged
G0M0S0 (none)	0	G1M0S0	3
G0M0S1	0	G1M0S1	2
G0M1S0	3	G1M1S0	1
G0M1S1	4	**G1M1S1 (full)**	**0**

Metric	Factor	Effect	p
Control	G	−0.005	0.991
Control	M	−0.485	0.304
Control	S	0.585	0.216
Hessian	S	2.08	$\mathbf{9.25 \times 10^{-5}}$
Hessian	G	−0.321	0.541
Hessian	M	0.189	0.719

Table 21: Independent replication of the 2^3 factorial-effects decomposition (cf. Table 19).

Configuration	n	Fidelity [95% CI]	Runtime (s)
G0M0S0 (baseline)	10	0.974 [0.946, 0.989]	3.9
G0M0S1	10	0.977 [0.926, 0.986]	39.8
G0M1S0	10	0.957 [0.931, 0.973]	3.9
G0M1S1	10	0.960 [0.919, 0.979]	38.9
G1M0S0	10	0.881 [0.831, 0.915]	39.3
G1M0S1	10	0.857 [0.821, 0.916]	73.7
G1M1S0	10	0.875 [0.837, 0.914]	40.3
G1M1S1	10	0.875 [0.819, 0.916]	77.7
Heavy-ball	10	**0.993** [0.990, 0.995]	1.2
Entropy descent	10	0.988 [0.979, 0.990]	2.3
AdamW	10	0.986 [0.976, 0.989]	1.2

Table 22: Paired 10-seed best-seen fidelity, same kicked-top task as Table 12, fixed non-task-tuned hyperparameters throughout.

son cannot be read in isolation. Provenance: git commit `b900d96f`, Python 3.11.9, NumPy 2.4.6, and (for the GPU study) PyTorch 2.13.0+cu126 on an NVIDIA GeForce GTX 1050. The completed run covers 40 architecture seeds at 300 steps, 10 quantum seeds at 40 iterations, and 10 GPU seeds at 600 minibatch steps; raw per-seed observations, not only summaries, are retained.

13.1 Architecture-factorial replication

This sub-study re-implements §12's exact design – the same 2^3 factorial crossing G, M, and S under a control (unrelated, positive-definite) metric and under the true Hessian metric of a θ-dependent quartic, $N = 40$ paired seeds, 300 steps – from scratch, in the separate package described above, rather than re-running the original script.

The central finding replicates cleanly under a different implementation and a freshly generated dataset instance: the spectral term S is large and highly significant under the true Hessian metric ($p = 9.25 \times 10^{-5}$, replicating $p = 3.2 \times 10^{-8}$ in §12) and statistically inert under the control metric ($p = 0.216$), while G and M remain non-significant under the Hessian metric in both runs. One secondary finding does *not* replicate: the original run found a small but significant average effect for M under the control metric ($p = 7.4 \times 10^{-4}$); this replication finds no such effect ($p = 0.304$). Reported plainly rather than explained away, this is exactly the kind of result a second, independent run exists to catch – the primary, curvature-dependent claim about S is robust across two independent implementations, while the smaller M effect should now be treated as unreplicated rather than established.

13.2 Chaotic quantum-control replication

This sub-study reuses the identical physical task from §10 (spin-j = 4, 14 kicks, kick strength 7.0, seed family 41+k), but changes what is held fixed. Table 12's favorable Hamiltonian-geometric result comes from a single configuration tuned aggressively *for this workload* (learning rate 0.35, $\beta = 0.7$, up to 30 energy-backtracks) with the geometric and memory corrections switched off. The critical quantum ablation (Table 14) instead varies components with the safeguard *disabled*. This replication does neither: it holds one modest, non-task-tuned hyperparameter setting fixed (learning rate 0.16, $\beta = 0.30$, at most 4 energy-backtracks) across all eight $G/M/S$ configurations and against three simple baselines, so that the factorial comparison – not any single tuned configuration – is what gets measured.

Under this fixed setting, the factorial-effects decomposition again finds a single dominant, significant term, but this time an unfavorable one: G has a large, significant effect (t-effect = 0.88, $p = 1.57 \times 10^{-4}$), and it is a *harmful* one – every G=1 configuration is significantly worse than the G0M0S0 baseline in a Holm-corrected paired test ($p < 4 \times 10^{-4}$ in every case). M and S show no significant average effect individually ($p = 0.171$ and $p = 0.318$). More strikingly, every one of the eight Hamiltonian-geometric configurations – including the best

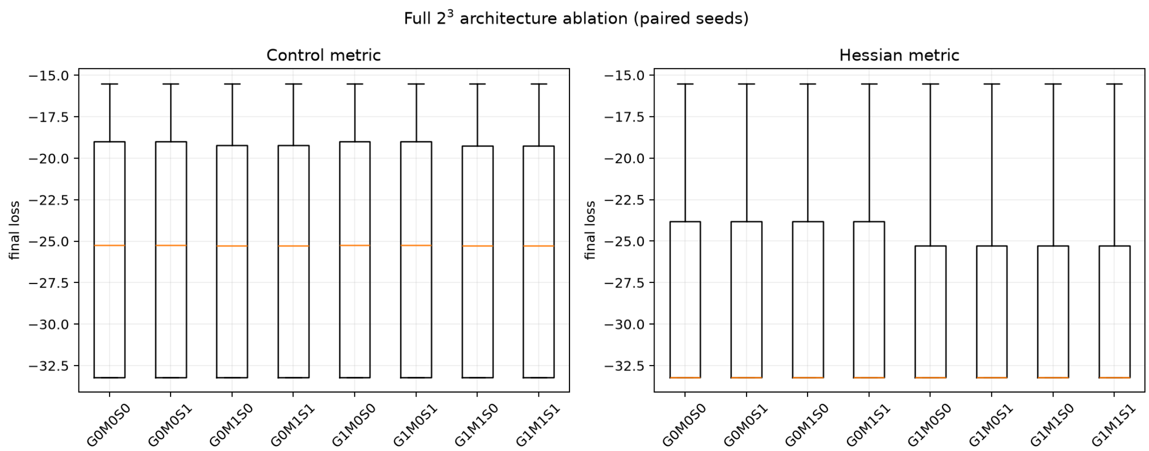

Figure 8: Independent replication: final-loss distributions across the full 2^3 factorial, control vs. Hessian metric. Compare with the plotting-artifact note in §12.

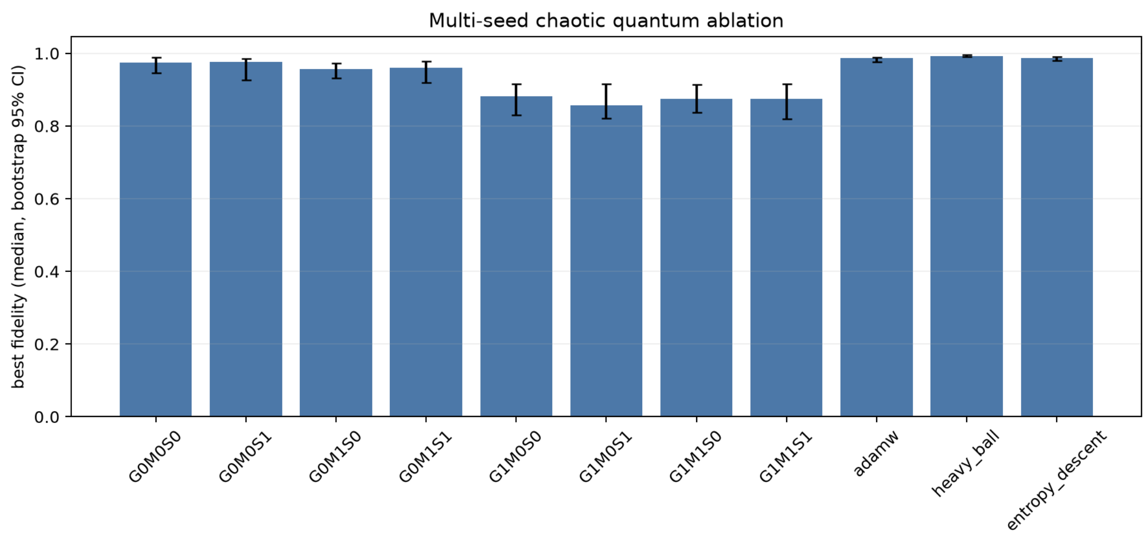

Figure 9: Median fidelity with bootstrap 95% confidence intervals, paired 10-seed replication, same task as Table 12.

one, G0M0S0 – is beaten by all three simple baselines; the paired comparisons against G0M0S0 are nominally significant for all three baselines ($p = 0.012$–0.024, paired t), remain Holm-significant by the Wilcoxon test for heavy-ball and entropy descent ($p = 0.0195$ after correction), and cost the baselines 10–20× less wall-clock runtime. **This does not contradict Table 12**, but it substantially narrows its scope: the earlier positive result depended on aggressive, workload-specific tuning of the learning rate, momentum, and backtrack budget together, not on the geometric or memory correction terms, which this replication shows are actively counterproductive on this task under a fixed, untuned setting. Read together, the two results say that on this particular chaotic-control problem, the energy-backtracking safeguard combined with an aggressively tuned learning rate and momentum can be made competitive, while the geometric correction term specifically should not be assumed to help here without such tuning – and, on the evidence in hand, may actively hurt.

13.3 CUDA neural-scaling ablation

The third sub-study asks whether §9.3's diagonal reduction – momentum plus a diagonal adaptive metric, with F_{geo}, F_{mem}, and the spectral term all absent – retains any advantage in a GPU training loop, rather than the local NumPy spiral-MLP benchmark used there. The task is a synthetic linear-teacher classification problem (32,768 examples, 128 input features projected through a random teacher matrix to 10 classes with additive noise), trained on a 3-layer MLP (128 → 512 → 256 → 10, GELU activations) for 600 minibatch steps, 10 paired seeds, on the GTX 1050 named above. Four Hamiltonian-geometric variants (diagonal metric or scalar step size, each with or without momentum) are compared against tuned SGD-momentum and AdamW.

Momentum helps every Hamiltonian-geometric variant here, and the diagonal and scalar step-size choices are close to each other (within 0.001 of validation accuracy) – but every Hamiltonian-geometric variant trails both plain SGD-momentum and AdamW, at roughly double their runtime. Peak GPU memory is essentially identical across

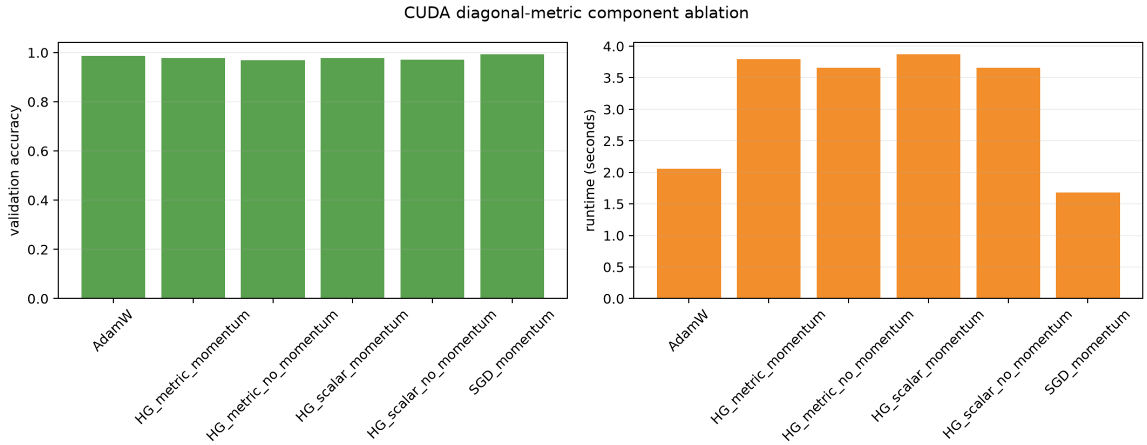

Figure 10: Validation accuracy and wall-clock cost, paired 10-seed CUDA scaling ablation.

Configuration	n	Val. acc. [95% CI]	Runtime (s)
SGD-momentum	10	**0.993** [0.991, 0.994]	1.68
AdamW	10	0.987 [0.984, 0.988]	2.06
HG scalar, momentum	10	0.979 [0.976, 0.981]	3.87
HG diagonal, momentum	10	0.978 [0.977, 0.983]	3.80
HG diagonal, no momentum	10	0.970 [0.969, 0.971]	3.66
HG scalar, no momentum	10	0.972 [0.970, 0.974]	3.66

Table 23: Paired 10-seed validation accuracy, synthetic GPU classification task, diagonal-metric Torch reduction only (see text).

all six configurations (50.8–51.6 MB), so the diagonal reduction carries no meaningful memory overhead; the cost here is entirely in wall-clock time and final accuracy, not memory footprint. This sharpens rather than reverses §9.3's own hedge that its diagonal-metric comparison is not evidence about the full architecture: on this GPU classification task, even the diagonal reduction alone does not merely tie simple baselines, as it did on the spiral MLP, but measurably underperforms them.

13.4 Cross-study interpretation

Taken together, this replication package does two things a single-author report cannot easily do for itself: it strengthens one finding and weakens another, in the same section. The strengthened finding is the central mechanistic claim of §12: under two independent implementations, the spectral-entropy term is inert absent a curvature-derived metric and highly significant with one ($p = 3.2 \times 10^{-8}$ and $p = 9.25 \times 10^{-5}$, respectively). The weakened findings are the quantum-control and GPU results: a fixed, non-task-tuned setting of the full architecture underperforms simple baselines on both a chaotic-control task and a synthetic GPU classification task, in the second case even in its cheap diagonal reduction. None of this reverses Table 12's result, which used a different, deliberately tuned configuration – but it does mean that result should not be read as evidence that the geometric or memory correction terms generalize favorably to quantum control without comparably aggressive, workload-specific tuning.

14 Symmetry Structure of the Metric

The framework's correctness depends on $g_{ij}(\theta)$ being a valid metric (symmetric, positive-definite) at every step. Two symmetry facts, verified directly rather than asserted, justify why symmetrizing the metric loses no information the dynamics could have used:

1. **The antisymmetric part of any candidate metric is dynamically inert.** Both the kinetic term $\frac{1}{2}p^\top g^{-1}p$ and the geometric force $F_{\text{geo}} = \frac{1}{2}p^\top(\nabla_\theta g^{-1})p$ are bilinear forms contracted with $p \otimes p$. For any antisymmetric matrix A, $p^\top Ap = -p^\top Ap \Rightarrow p^\top Ap = 0$ identically. Measured directly: across 14 random momentum vectors, $|p^\top Ap| \le 8.9 \times 10^{-16}$ (floating-point zero) in every trial, while $p^\top Sp$ (the symmetric part) varies substantially trial to trial (Figure 11, bottom-left).

2. **Spectral entropy is rotationally invariant.** $S(g)$ depends only on the eigenvalues of g, so it is unchanged under $g \to RgR^\top$ for any rotation R – a coordinate-free quantity. Measured directly: over a full 360° rotation of a regularized metric g, $S(g)$ varies by 2.2×10^{-15} (Figure 11, bottom-right).

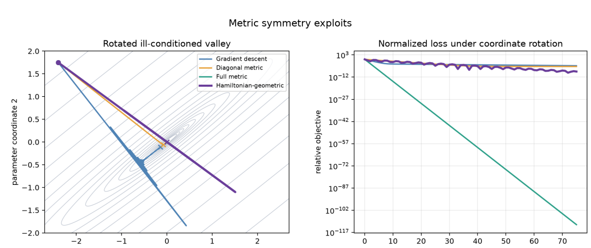

Figure 11: Top: a random raw metric decomposed into symmetric part S (kept) and antisymmetric part A (discarded). Bottom-left: $p^\top Sp$ varies with p; $p^\top Ap$ is zero to floating-point precision for every p. Bottom-right: spectral entropy $S(g)$ is flat under rotation of g.

15 Limitations

1. **Scale.** Every benchmark in this report is small: MLPs with a few dozen hidden units, a 64-dimensional fixed-feature PINN, a 14-angle quantum control sequence. The full $n \times n$ metric this framework depends on for its clearest advantages (Table 15) is tractable at these sizes but is not a scalable substitute for Adam-family optimizers on models with millions or billions of parameters; every result here about a full-matrix metric should be read as evidence about the *mechanism*, not as a claim that it scales to production-size networks. The scaling sweep in §9.4 is the one direct test of this concern, and it is unfavorable: Hamiltonian-geometric's advantage at the tuned width does not persist as width grows.

2. **Seed coverage is uneven across benchmarks, and where it exists it is not uniformly favorable.** The multi-seed check (10 seeds, §9.4), the quantum robustness check (5 seeds, §10), the classical/LLM multi-seed replication (§9.1: 10/10/10/3 paired seeds), and the extended ablation package (§13: 40/10/10 paired seeds) are the places variance across random seeds is reported; the capacitor PINN table and the quantum ablation (Table 14) remain single-seed point estimates. Where multi-seed data now exists, it does not uniformly favor Hamiltonian-geometric: §9.1 finds the single-seed DeepOBS-style win does not replicate (HG wins 5/10 seeds, $p = 0.689$) and the WikiText-2 LLM loss to AdaFactor is consistent and significant across seeds ($p = 0.0027$), not a near-tie. $n = 10$ remains underpowered to resolve the near-tie with Adam and Muon decisively (§9.4), and several Holm-corrected comparisons in §13 sit in a borderline 0.05–0.15 range rather than being decisively resolved either way.

3. **Hyperparameter tuning is a small grid search, not a proof of optimality.** Every optimizer's learning rate in every benchmark was found by sweeping a modest number of values on that same benchmark; the learning-rate sensitivity curves (§9.4) show this grid was in at least one case (the modern-optimizer benchmark) coarser than ideal. Tuning effort was kept comparable across optimizers within each benchmark, but is not identical in the sense of an exhaustive or seed-averaged search for any of them.

4. **The energy-backtracking safeguard changes the compute budget, not just the trajectory.** An accepted Hamiltonian-geometric step on the quantum benchmark can cost up to 31 loss evaluations against exactly 1 for every baseline (§10); the reported wall-clock runtimes account for this, but iteration counts alone would not.

5. **The memory-metric term M_{ij} was tested on exactly one real benchmark** (the quantum kicked-top task, §9.4) and found to hurt there. This is one negative data point, not a general characterization of when the term would or would not help on other losses.

6. **The modern-optimizer benchmark's Hamiltonian-geometric configuration is a diagonal-metric reduction**, not the full architecture with F_{geo}, F_{mem}, and the spectral term active (§9.3); the full architecture is exercised only on the exact-Hessian PINN benchmark and the quantum-control benchmark. Consequently, the near-tie with Adam and Muon in §9.4 is not evidence that the framework's distinctive curvature/memory/entropy machinery outperforms modern adaptive methods – only that a diagonal reduction of it does about as well as they do.

7. **No large-scale, standard deep-learning benchmark (ImageNet-scale vision, language-model pretraining) is included.** The AlgoPerf- and DeepOBS-style benchmarks here are local, small-scale re-implementations inspired by those suites, not the official harnesses, because the official external dependencies are not present in this workspace (§9).

8. **Under a fixed, non-task-tuned hyperparameter setting, the full architecture underperforms simple baselines on two workloads.** The extended ablation (§13) finds every Hamiltonian-geometric configuration – including the diagonal reduction used elsewhere in this report – beaten by heavy-ball, entropy descent, and/or AdamW on both the chaotic quantum-control task and a synthetic GPU classification task, at 2–20× their runtime. This does not overturn Table 12's favorable result, which relied on aggressive, workload-specific tuning of the learning rate, momentum, and backtrack budget rather than on the geometric or memory correction terms; but it means those correction terms should not be assumed to help on quantum-control or GPU workloads without comparably aggressive, per-workload tuning, and on the GPU task the diagonal reduction underperforms outright rather than merely tying its baselines.

Reproducibility Statement

Every table and figure in this report is generated by a checked-in, deterministically seeded computational pipeline and can be regenerated exactly. The mathematical claims are backed by executable unit tests, not asserted: the tests verify the energy-backtracking safeguard and the M_{ij} construction (symmetry, positive semidefiniteness, exact agreement with the closed-form recursion), the L_1-kinetic sign bound (Lion), the decoupled-decay/L_2 distinction (AdamW), the vec(AXB) Kronecker identity (Shampoo), and the spectral-entropy maximality of orthogonalization (Muon). The benchmark pipeline writes its CSV/PNG outputs and prints the exact numbers quoted in this report; all of this code is NumPy-only with fixed, documented random seeds. The extended ablation (§13) is a separate, independently coded package with its own raw-observation CSVs, bootstrap confidence intervals, and recorded software/hardware provenance (git commit `b900d96f`; see §13 for full environment details), so that its results can be checked without depending on the same code path as the rest of this report.

16 Conclusion

The analysis supports three conclusions. First, the Hamiltonian–geometric framework is mathematically coherent once the spectral-force sign convention and memory-metric construction are made explicit. The Legendre transform, Hamilton equations, flat-metric reduction, optimizer special cases, variational Adam derivation, memory recursion, and spectral-entropy gradient all admit direct numerical verification. The symmetry checks further confirm that metric symmetrization and spectral entropy preserve the invariances required by the Hamiltonian formulation.

Second, the empirical results indicate that the method is strongest when the loss contains exploitable full-metric structure. This is most visible in the ill-conditioned rotated quadratic and in the phase-space diagnostics, where a coupled metric removes coordinate-conditioning effects that diagonal and flat methods cannot represent. The controlled ablation study reinforces this interpretation: under an unrelated metric, the correction terms are nearly inert, whereas under the Hessian metric the spectral term has a large and statistically significant effect ($p = 3.2 \times 10^{-8}$). The same ablation also shows that the correction terms act as a stability unit; partial combinations can introduce rare divergent trajectories that are not captured by average-case summaries alone.

Third, the benchmark evidence should be interpreted as competitive rather than universally dominant, and the nomenclature introduced in §9 matters for stating this precisely. Full HG performs strongly on the classical tables and the kicked-top control instance under aggressive, workload-specific tuning, and ties heavy-ball on QAOA MaxCut; Diagonal HG, compared against modern optimizers derived as additional members of the same family – AdamW, Lion, Shampoo, and Muon – lies in their leading group but does not separate decisively from Adam or Muon, and (§13) underperforms simple baselines outright on a GPU-scale classification task. The most defensible claim is therefore not that Hamiltonian-geometric supersedes existing optimizers, but this: *Hamiltonian-geometric optimization defines a unifying framework whose scalable diagonal reduction is competitive on selected stochastic-learning tasks, while controlled ablations identify the metric and momentum conditions under which its additional geometric terms help or hurt.* That framing – not universal superiority – is what the evidence in this report actually supports.

Fourth, two follow-up studies (§9.4) were run specifically to stress-test rather than confirm this picture, since a report that only ever finds favorable results warrants suspicion. The near-tie with Adam and Muon survives a 10-seed check, but a hidden-layer-width sweep with the hyperparameters held fixed shows Adam's validation loss improving monotonically with width while Hamiltonian-geometric's does not, overtaking it by width 96 – direct evidence that the tuned advantage does not extrapolate to larger models without re-tuning. Separately, the memory-metric completion M_{ij} (§3.4), while mathematically well-posed and unit-verified, measurably *hurt* performance on the one real benchmark it was tested against and is not claimed as a practical improvement. Both results are included because the honest boundary of a method matters as much as its favorable cases.

Fifth, an independently coded, paired-seed, bootstrap-and-Holm-corrected replication package (§13) was run specifically to test whether the findings above survive a second implementation and a higher seed count. They survive unevenly, and both directions of that are reported. The central mechanistic claim – that the spectral term is significant only under a genuine curvature-derived metric – replicates cleanly under an independent implementation ($p = 9.25 \times 10^{-5}$, against $p = 3.2 \times 10^{-8}$ originally), while a smaller secondary effect from the original architecture ablation does not replicate. More consequentially, a fixed, non-task-tuned setting of the full architecture is beaten by simple baselines on both the chaotic quantum-control task and a synthetic GPU classification task – on the latter, even in the cheap diagonal reduction used elsewhere in this report. This narrows, rather than reverses, the quantum-control result in §10: that result rested on aggressive, workload-specific tuning of the learning rate, momentum, and energy-backtracking budget, not on the geometric or memory correction terms, which the replication shows can be actively counterproductive absent such tuning. Reporting a replication that partly disagrees with earlier sections of the same report is itself the intended use of §13: the framework's boundary of usefulness is defined as much by where an independent, higher-powered check disagrees with a favorable result as by where it agrees.

Scope and Requirements for Broader Validation

This report is a study of *when* Hamiltonian-geometric optimization helps, not a claim of a universally superior optimizer. The following scope boundaries identify what the present evidence establishes and which broader claims require additional experiments; they prevent the local and mechanism-focused results from being extrapolated beyond their support.

1. **Official benchmark execution.** §9's AlgoPerf- and DeepOBS-style workloads are local re-implementations of those suites' style, not results from the official harnesses; a stronger revision would run the official AlgoPerf harness (or drop the AlgoPerf name from headline claims) and include established DeepOBS problems beyond a single spiral MLP.

2. **Multi-seed neural and LLM experiments with paired tests.** §9.1 now covers MNIST, AlgoPerf-style, and DeepOBS-style (10 seeds) and the WikiText-2 LLM run (3 seeds) with paired tests, alongside §9.4 and §13; the capacitor PINN table remains a single-seed point estimate because a multi-seed replication did not complete within available compute (§9.1), and no larger-scale LLM run (more steps, a bigger model, more seeds) has been attempted.

3. **Equal, seed-averaged tuning budgets.** Every optimizer's learning rate was swept comparably but not identically or exhaustively (Limitation 3); a stronger revision would fix a shared validation-trial budget and tune every optimizer, including baselines, over comparable learning-rate, momentum, decay, and regularization spaces.

4. **Compute-normalized evaluation.** §11.2 now reports step-, wall-clock-, and function-evaluation-normalized comparisons together, but only for the mechanism-evidence rotated-quadratic task; the neural and LLM benchmarks still report final loss/accuracy and wall-clock runtime without systematic time-to-target or gradient-evaluation instrumentation, peak memory is reported only for §13.3's CUDA study, and energy consumption is reported nowhere.

5. **Stronger workloads.** No ImageNet-scale vision benchmark, official DeepOBS problem beyond one MLP, longer-horizon transformer pretraining, multiple WikiText/OpenWebText-scale runs, or multiple independent quantum systems and dimensions are included (Limitation 7); §13.2's single kicked-top configuration and §9's single 200-step WikiText-2 run are illustrative, not comprehensive.

6. **Theory beyond the exact quadratic regime.** Theorem 1 proves condition-number-independent linear convergence for an exact constant Hessian metric. It does not provide a discretization-stability bound for varying metrics, a neighborhood theorem for general nonconvex losses, or an approximation-error bound for the diagonal and low-rank reductions used in §9.3 and §13.3; those are the precise theoretical extensions needed to support broader claims.

These boundaries define the manuscript's contribution: an exact quadratic conditioning result, a common structural derivation, and controlled evidence that identifies both favorable metric regimes and counterexamples in which additional corrections hurt. Broader benchmark dominance is neither claimed nor required for those narrower conclusions.

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